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Interpretability and Conditional Generation with Flow-Matching Models

- Presentation for the degree of M.Sc. in Computer Science
- Task assigned by: Prof. Dr. Björn Ommer
- Supervised by: Johannes Schusterbauer, M.Sc., PhD Student, Ming Gui, M.Sc., PhD Student
- Date of presentation: 09.09.2025

Motivation

The purpose of this work is to empirically explore if it is possible to utilise **representation learning** methods to improve **model interpretability** and with-it **generative control** in state-of-the-art deep generative models such as Flow Matching [1-3].

Motivation and Research Gap:

- **Trending research topic:** an increasing amount of work is recently coming out (**Dec. 2024 - 2025**)
  - Latent noise spaces (e.g., *Golden Noise* [4], *NoiseRefine* [5])
  - Frequency-aware modelling (e.g., *DDT* [6], *Diffusion as spectral autoregression* [7], *Guidance in frequency domain* [8])
- **Self-guidance** is a new research area with growing interest
  (*e.g., utilising k-Means clustering for self-labelling* [9] *or learned internal model features directly* [6]).

Research Objectives

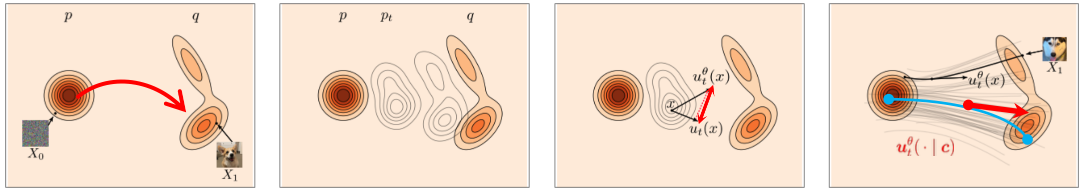
1. **RQ1:** Do the temporally distributed, noisy latent representations along the continuous probability paths of Flow Matching models encode *semantically meaningful information* (e.g., texture, pose, illumination) that can be readily extracted through a self-supervised representation learner?

2. **RQ2:** Can we learn an auxiliary latent space which exhibits desirable properties such as *smoothness* (i.e., coherent, attribute-consistent transformations) and *disentanglement* (i.e., a single latent unit corresponds only to changes in a single generative factor), to enable more human-interpretable generative processes?

3. **RQ3:** Can the extracted features of this space be leveraged as *self-guidance* signals to enable more fine-grained generative control, and semantic consistency with the target distribution, whilst maintaining output fidelity?

Background

Conditional Flow Matching



(a) Data. **(b)** Path design. **(c)** Training. **(d)** Sampling.

- Transforms samples $\boldsymbol{x}_0$ from a **known distribution** to samples $\boldsymbol{x}_1$ from an **unknown target distribution** [8-10].
- Conditional Flow Matching aims to learn the parameters of a conditional velocity field $\boldsymbol{v}_\theta$, describing how a point moves over time:

Step 1: Path design – design a conditional probability path p_t interpolating between q and p (Figure b)

➢ Gaussian source distribution $p := p_0 = \mathcal{N}(\boldsymbol{x} \mid 0, I)$

➢ Forward time process (or linear interpolation) conditioned on a target example $\boldsymbol{X}_1 = \boldsymbol{x}_1$

$$\boxed{X_{t|1} = tx_1 + (1-t)X_0} \sim p_{t|1}(\cdot \mid x_1) = \mathcal{N}\big(\cdot \mid tx_1, (1-t)^2 I\big)$$

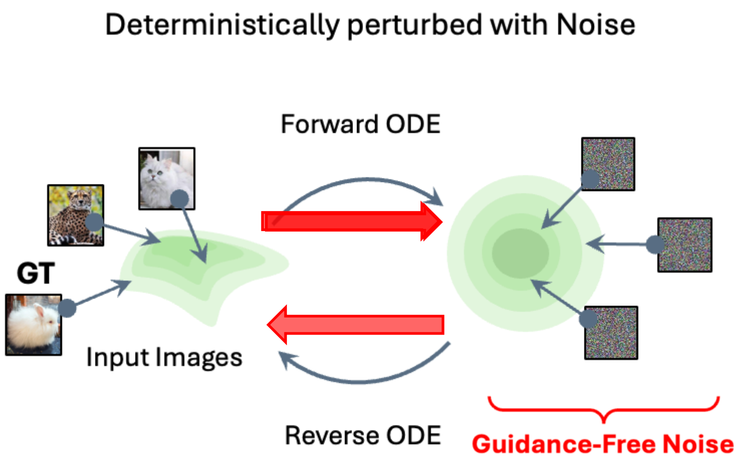
With conditional velocity field $\boxed{u_t(x \mid x_1) = \dfrac{x_1 - x}{1-t}}$ when solving $\dfrac{d}{dt} X_{t|1} = u_t(X_{t|1} \mid x_1)$

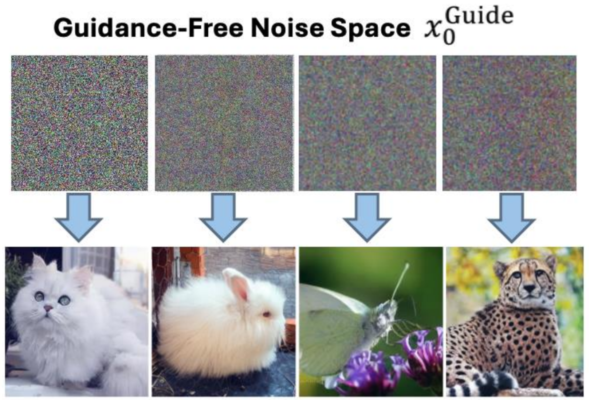
Step 2: Training – trains a velocity field generating p_t using regression (Figure c)

➢ Loss becomes tractable under conditional probability paths

$$L_{\mathrm{CFM}}(\theta) = \boxed{\mathbb{E}_{t,X_0,X_1} \left\| u_\theta^t(X_t) - (X_1 - X_0) \right\|^2}, \quad t \sim U[0,1],\ X_0 \sim \mathcal{N}(0, I),\ X_1 \sim q.$$

Guidance-Free Noise Space





(b) Images generated from samples with applied deterministic noise - **without guidance.**

Guidance-free Noise

Under Flow Matching, the forward ODE encodes images into unique initial noise samples $\boldsymbol{x}_t$ that are **"guidance-free"** [5].

- ➢ Samples retain necessary high-level **semantic information** about their input (e.g., colour, shape, location, object identity, silhouette) which allows almost full reconstruction (i.e., **cycle-consistency**) [8, 13, 15].
- ➢ Enables high-quality image synthesis **without additional guidance.**

Guidance-free Dataset

- Run the forward-time mapping from a latent image to a corresponding, **unique guidance-free noise** sample in guidance-free noise space.
- Subsample for temporal coverage

$$t \in \{0.7, \ldots, 0.5, 0.2, 0.0\}$$

Frequency Analysis

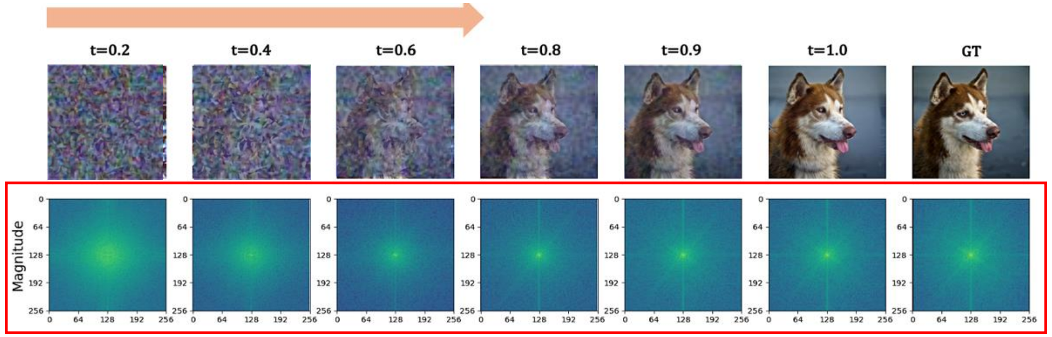


Figure: Image spectra in frequency domain. Selected sample from ImageNet along with its magnitude spectra (bottom).

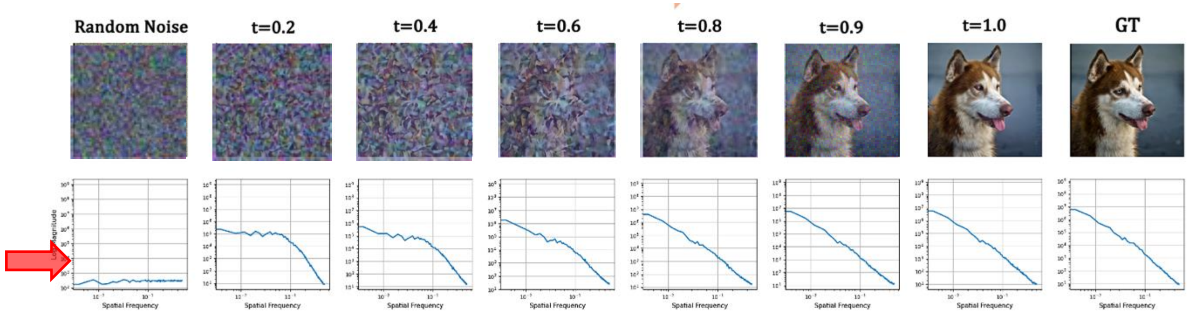


Figure: Image magnitude spectra in frequency domain. Selected sample from ImageNet along with its magnitude spectra in log-scale (bottom).

Noise Image Spectra

- **Image spectra:** decomposing an image into its constituent **spatial frequency** components to visualise what content is degraded or retained in frequency domain [6-7].
  - ➢ **Centre:** low-frequencies (e.g., global structure, surfaces, background, colour distribution)
  - ➢ **Outward edges:** high-frequencies (e.g. texture, hair strains, edges, object boundaries)
- **Reverse denoising**: Coarse-to-fine iterative refinement approach where we perform a single step of refinement at a time.
  - ➢ The higher the noise level, the larger the scale of features captured and vice versa

Representation Learning Framework

Desirable Properties

Representation Learning
- Decodable and semantically meaningful latents
- Disentangled and smooth latent space

Image Synthesis
- Steerable generation
- High-fidelity generation
- Fast inference

Hybrid Framework of Two Complementary Methods:

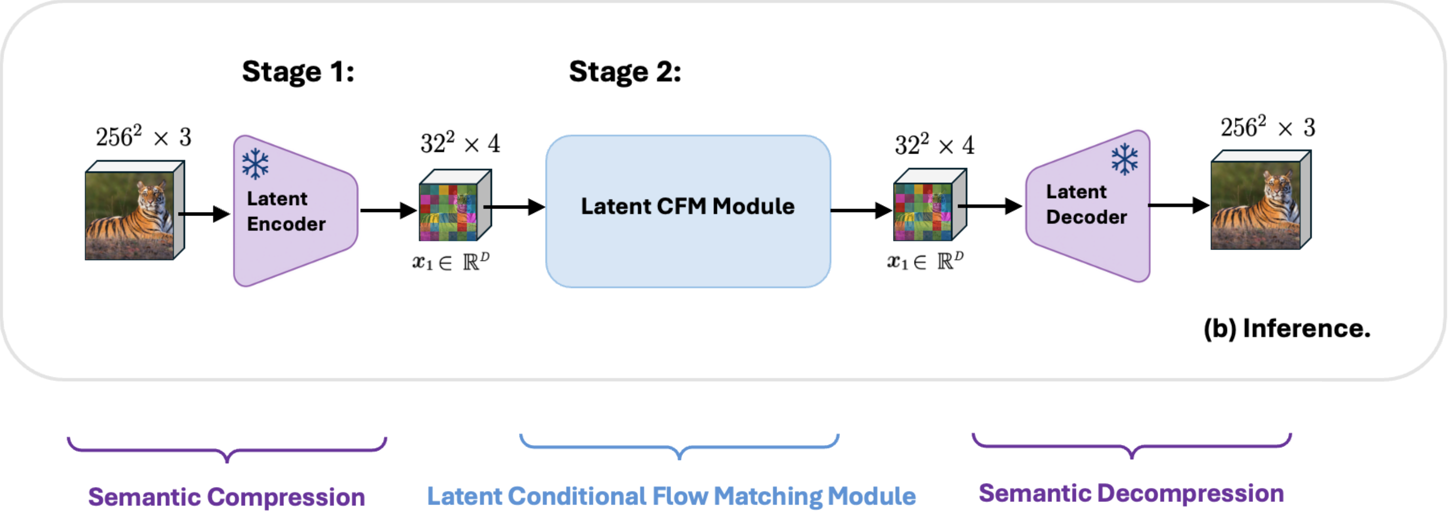
Flow Matching:

- ✓ High-fidelity generation
- ✓ Low-curvature paths
- ✓ Fast sampling
- ✗ No dedicated component for representation learning
- ✗ Temporally distributed, high-dimensional latent space
- ✗ Lower interpretability

ß-Variational Autoencoder:

- ✓ Representation learning
- ✓ Self-supervised
- ✓ Disentanglement
- ✓ Single-step sampling
- ✗ Inferior generative power, L2 loss can lead to global blurriness

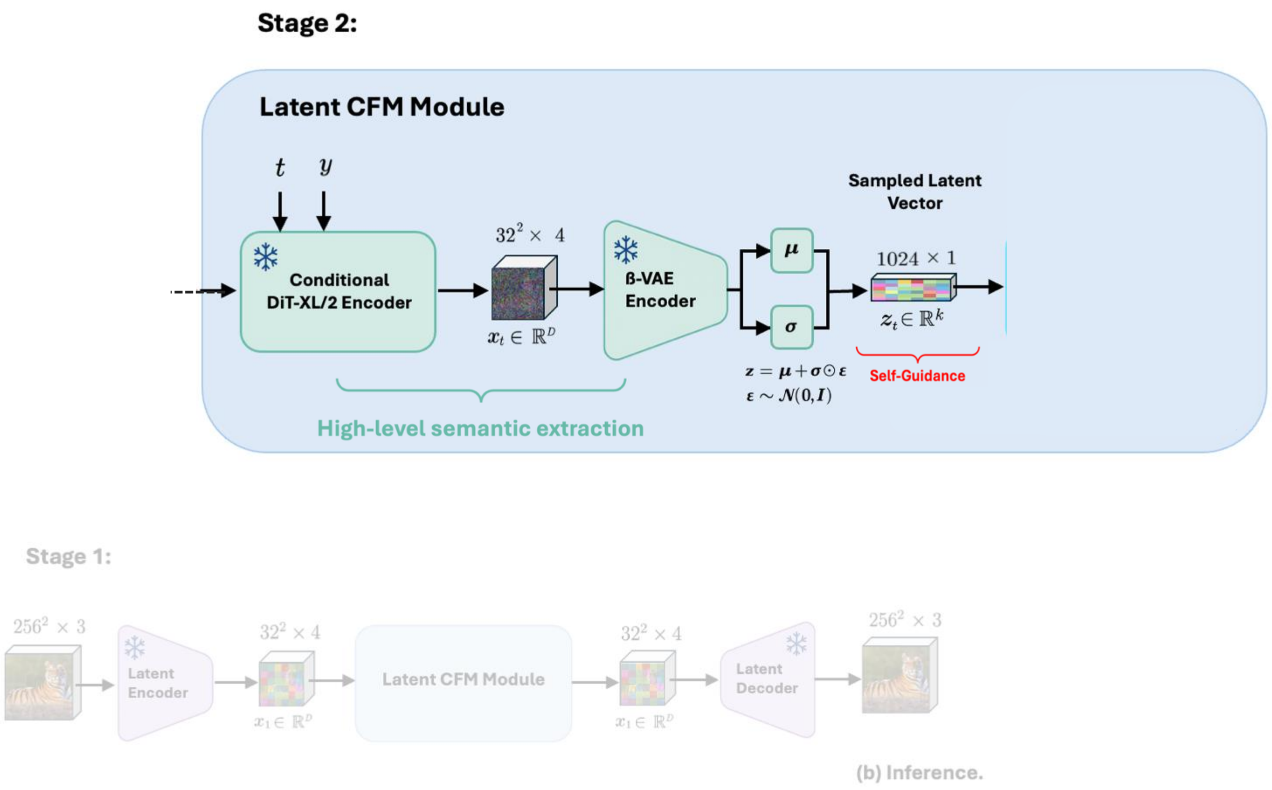
Representation Learning Framework



Two-stage Latent Framework

- **Stage 1:** high-level latent representation is extracted through a fixed latent encoder-decoder network [22].
- **Stage 2:** Flow Matching and ß-VAE processes operate in the fixed representational space allowing abstract feature information extraction and detailed appearance recovery.

Representation Learning Framework



Decoupled Hybrid Design

High-level Semantic Extraction

- **DiT-XL/2 encoder** embeds learned high-level and low-level semantic features into generated latent noise codes x_t over timesteps
- **ß-Variational Autoencoder** aggregates high-level semantic features from noise codes and compresses them into a lower-dimensional, non-spatial, smooth vector code z_t **(self-guidance)**

Low-level Stochastic Reconstruction

- **DiT-XL/2 decoder** is guided by high-level semantic codes z_t **(self-guidance)** to map noise back to the image manifold whilst recovering missing low-level information [19-20].

Linear Interpolation

Stage 2:

Algorithm 1 Latent Space Linear Interpolation (LERP)

Require: Two latent noise samples $\boldsymbol{x}_t^{(1)}, \boldsymbol{x}_t^{(2)}$ at timestep t, encoder $\mathcal{E}_\phi$, number of interpolation steps N

Ensure: Interpolated latent codes $\{\boldsymbol{z}_t^{(\lambda)}\}$

1: Encode inputs to $\mathcal{W}$ latent space:

$$\boldsymbol{z}_t^{(1)} \leftarrow \mathcal{E}_\phi(\boldsymbol{x}_t^{(1)}), \quad \boldsymbol{z}_t^{(2)} \leftarrow \mathcal{E}_\phi(\boldsymbol{x}_t^{(2)})$$

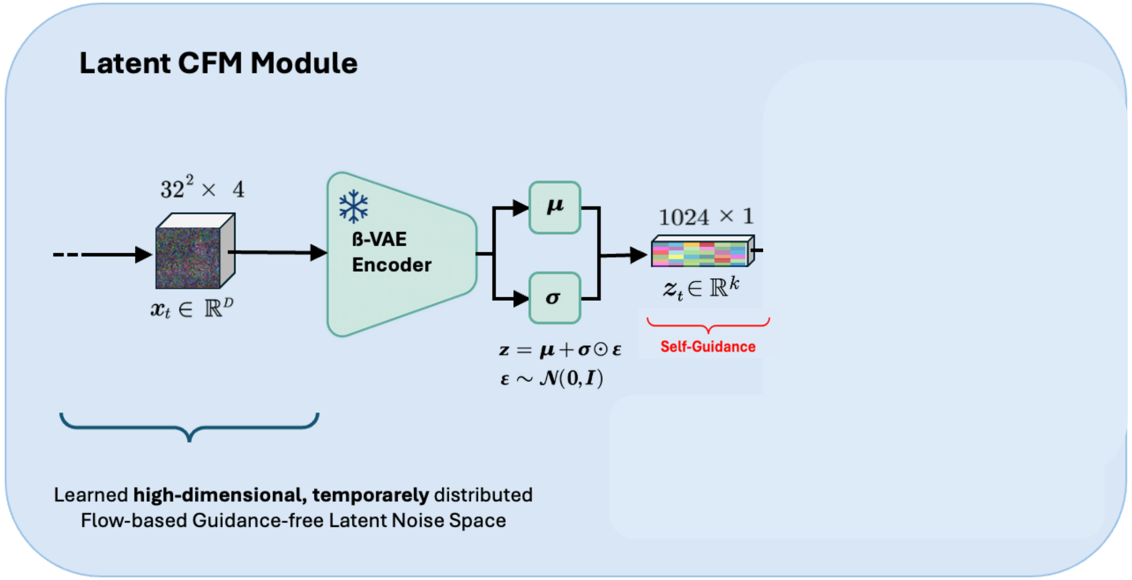
2: **for** $i = 0$ to $N-1$ **do**

3: Compute interpolation factor: $\lambda \leftarrow \frac{i}{N-1}$

4: Linear interpolate in latent space:

$$\underbrace{\boldsymbol{z}_t^{(\lambda)}}_{\text{Self-Guidance}} \leftarrow (1-\lambda) \cdot \boldsymbol{z}_t^{(1)} + \lambda \cdot \boldsymbol{z}_t^{(2)}$$

5: **end for**



Ablation

Evaluation of Reconstruction Output Fidelity / Bayesian Latent Space Smoothness

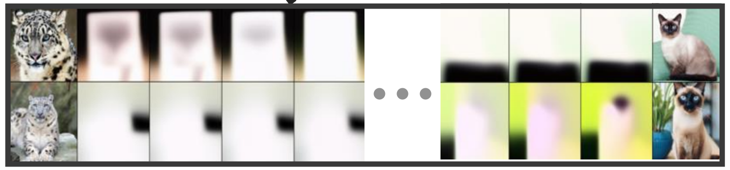
Reconstruction Task on Output Fidelity

Task ($t_{\text{source}} \to t_{\text{target}}$)	PSNR ↑	MSE ↓	MAE ↓	CosSim ↑
0.00 → 0.00	17.41	0.0182	0.1077	0.9690
0.20 → 0.20	18.21	0.0127	0.0935	0.9782
0.50 → 0.50*	**19.73**	**0.0108**	**0.0826**	**0.9830**

Table: The effect of decreasing noise-levels under a reconstruction task on output fidelity. * marks the baseline.

Reconstruction Task on Latent Space Smoothness

Task ($t_{\text{source}} \to t_{\text{target}}$)	L-PPL ↓	L-ISTD ↓
0.00 → 0.00	**0.218**	**0.054**
0.20 → 0.20	0.420	0.087
0.50 → 0.50*	0.846	0.100



Visualisation of interpolation in ß-VAE latent space under increased input noise-levels.

Task 1: Reconstruction

- Evaluation over ß-VAE models trained to reconstruct their noisy inputs in latent space.

$t_{\text{source}} = t_{\text{target}}$ with fixed $\beta = 0.1$

- **Highest performance** at medium noise (0.50 → 0.50)
- **Lowest performance** at full noise levels (0.00 → 0.00)
  reconstructing high levels of stochasticity is most challenging

- **Smoothest latent trajectory** at full noise levels (0.00 → 0.00)
- High noise smooths input signal and may cause collapse or trivial solutions. Subregions in manifold might lose meaning.

Ablation

Evaluation of Denoising Output Fidelity / Bayesian Latent Space Smoothness

Denoising Task Output Fidelity under Decreasing Noise-Levels.

Task ($t_{\text{source}} \to t_{\text{target}}$)	PSNR ↑	MSE ↓	MAE ↓	CosSim ↑
0.00 → 1.00	17.41	0.0182	0.1077	0.9690
0.20 → 1.00	17.24	0.0189	0.1096	0.9570
0.50 → 1.00	**20.94**	**0.00995**	**0.0793**	**0.9810**

Denoising Task on Latent Space Smoothness

Task ($t_{\text{source}} \to t_{\text{target}}$)	L-PPL ↓	L-ISTD ↓
0.00 → 1.00	**0.346**	**0.060**
0.20 → 1.00	0.808	0.112
0.50 → 1.00	0.571	0.119

Table: The effect of decreasing noise-levels under a reconstruction task on latent space smoothness. Values are adjusted by 1e-4 for readability.

Task 2: Denoising

- Evaluation over ß-VAE models trained to project the noisy inputs back to the clean image manifold [12-13].

  $t_{\text{target}} = 1.00$ with fixed $\beta = 0.1$

- **Highest performance** at medium noise (0.50 → 1.00)
- **Lowest performance** at full noise levels (0.00 → 1.00)
  denoising by learning to map noisy sample back to latent image manifold is most challenging

- **Smoothest interpolation path** at full noise levels (0.00 → 1.00)
- High noise smooths input signal and may cause collapse or trivial solutions. Subregions in manifold might lose semantic meaning.

Ablation

Evaluation of increasing β Regularisation Strength on Reconstruction Fidelity / Smoothness

The Effect of Regularisation Pressure on Denoising Accuracy

β	PSNR ↑	MSE ↓	MAE ↓	CosSim ↑
1e-4	**17.75**	**0.0118**	**0.1031**	**0.9694**
0.1	17.24	0.0189	0.1096	0.9568
0.5	16.59	0.0270	0.1318	0.9542
1.0†	17.57	0.0175	0.1052	0.9648
2.0	17.45	0.0127	0.1190	0.9606
3.0	17.18	0.0192	0.1105	0.9670
5.0	14.99	0.0317	0.1564	0.9808

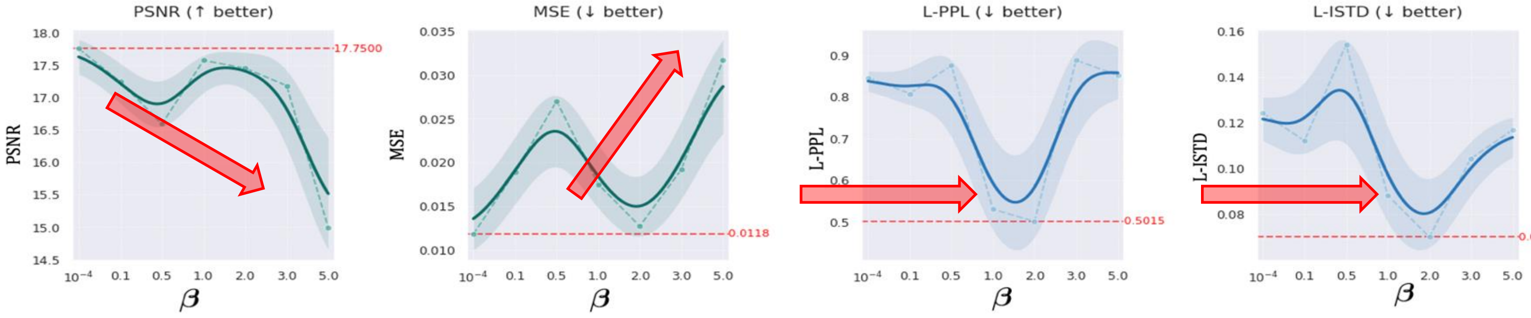
Table: The effect of increasing ß-hyperparameter under a denoising task on output fidelity. † marks the standard VAE model.

The Effect of Regularisation Pressure on Latent Space Smoothness

β	L-PPL ↓	L-ISTD ↓
1e−4	0.8463	0.1243
0.1	0.8077	0.1122
0.5	0.8753	0.1542
1.0†	0.5305	0.0884
2.0	**0.5015**	**0.0702**
3.0	0.8878	0.1044
5.0	0.8515	0.1169

Table: The effect of increasing ß-hyperparameter under a denoising task on latent space smoothness. † marks the standard VAE model. Sclaed by 1e-4 for readability.

The Effect of Regularisation Pressure on Reconstruction Fidelity and Latent Space Smoothness



ß-Hyperparameter

- Evaluation over Bayesian encoder trained on strongly corrupted samples with fixed target timestep.

  $$t_{\text{source}} = 0.20 \rightarrow t_{\text{target}} = 1.00$$

- **Disentanglement-Fidelity Trade-off:**
  - Lower values encourage capturing of more **fine-grained** information for improved filling in of missing information.
  - Higher values incure **loss of information** due to capacity constraints but increase disentanglement.
- **U-Shape Trend**: Very low and very high bottleneck pressures reduce smoothness.

Qualitative Results

Selected Samples Generated using the Self-Conditioned DiT-XL/2 Velocity Decoder

Ground-truth and Synthesized Samples for DiT-XL/2 with self-guidance CFG under weight $w_1 = 1.0$

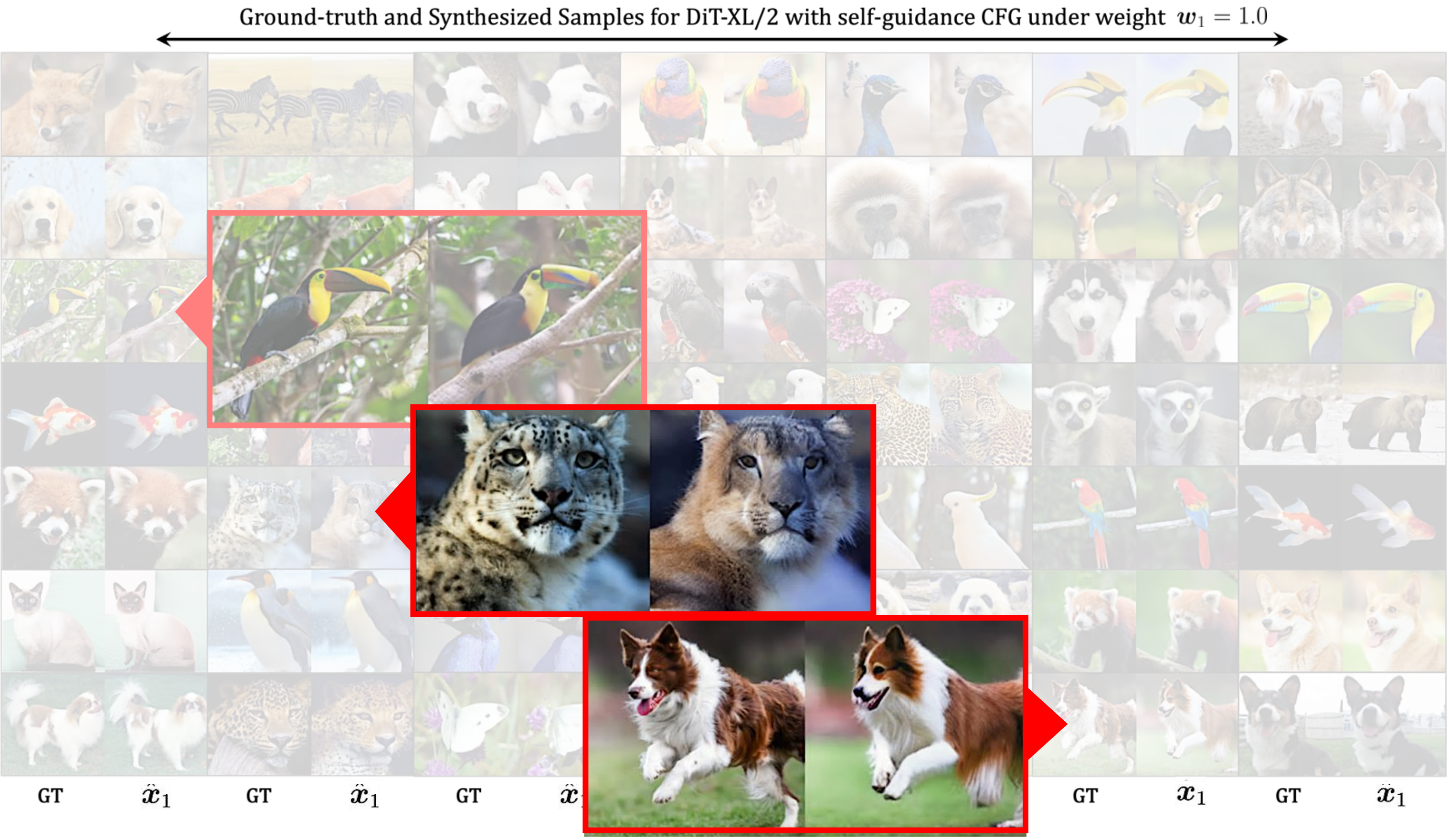
GT $\hat{x}_1$ GT $\hat{x}_1$ GT $\hat{x}_1$ GT $\hat{x}_1$ GT $\hat{x}_1$ GT $\hat{x}_1$ GT $\hat{x}_1$

Self-Guidance Conditioning

- Extracted Self-Guidance code from ß-VAE encoder trained with fixed $\boldsymbol{\beta} = 0.1$
- Using classifier-free guidance weight $\boldsymbol{w}_1 = 1.0$
- **High-fidelity reconstructions** maintaining layout, colour distribution, lighting, location.

Qualitative Results

Selected Samples Generated using the Self-Conditioned DiT-XL/2 Velocity Decoder



Self-Guidance Conditioning

- Extracted Self-Guidance code from ß-VAE encoder trained with fixed $\boldsymbol{\beta} = 0.1$
- Using classifier-free guidance weight $\boldsymbol{w}_1 = 1.0$
- **High-fidelity reconstructions** maintaining layout, colour distribution, lighting, location.
- **Limitation:** Reduced accuracy over very small objects, fine-grained stochastic background details (e.g. leaves) and distinct object patterns (e.g. leopard spots, fish scales, beak colour).

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Self-Guidance

Algorithm 1 Self-Guidance for Semantic Control

Require: noisy input x_t, timestep t, optional class label y
Ensure: velocity field $v_{\mathrm{CFG}}(x_t, t)$

1: Encode noisy input with Bayesian encoder: $z_t \sim q_\phi(z_t \mid x_t)$

2: Apply stochastic conditioning dropout:

$$\tilde{z}_t = \begin{cases} z_\emptyset & \text{if } r \leq p_{\text{drop}}, \\ z_t & \text{otherwise,} \end{cases} \quad r \sim U(0,1)$$

3: Train two velocity predictors:

$$v_\theta^{\text{cond}}(x_t, t; z_t), \quad v_\theta^{\text{uncond}}(x_t, t; z_\emptyset)$$

4: Combine velocity fields during inference:

$$\begin{aligned} v_{\mathrm{CFG}}(x_t, t) = & \boxed{v_\theta(x_t, t; z_\emptyset, y_\emptyset)} \\ & \boxed{+ \omega_z \Big(v_\theta(x_t, t; z_t, y_\emptyset) - v_\theta(x_t, t; z_\emptyset, y_\emptyset) \Big)} \\ & \boxed{+ \omega_y \Big(v_\theta(x_t, t; z_\emptyset, y) - v_\theta(x_t, t; z_\emptyset, y_\emptyset) \Big)} \end{aligned}$$

Unconditional

Self-Guidance

Class-Conditional

5: Integrate probability flow ODE:

$$p_0 = \mathcal{N}(0, I) \xrightarrow{v_{\mathrm{CFG}}(x_t, t)} p_1 = p_{\text{data}}$$

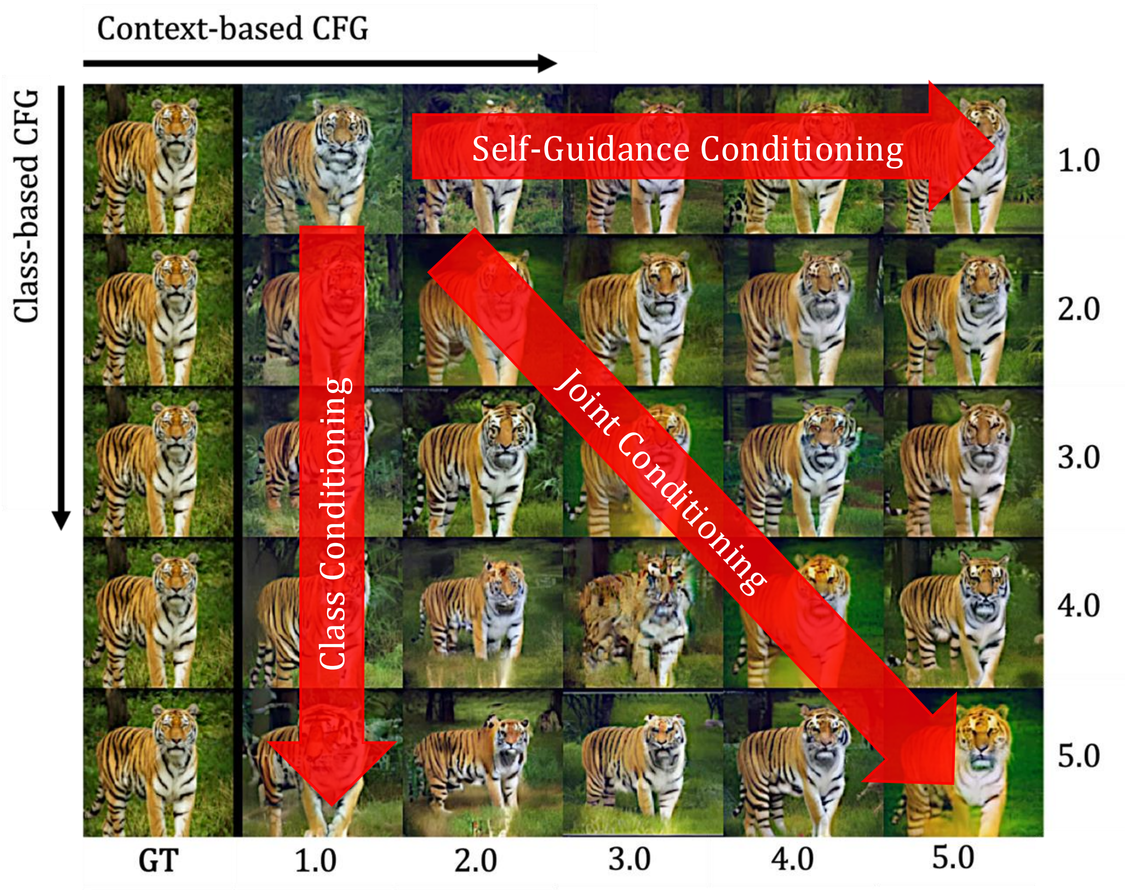
Self-Guidance Algorithm

- **Self-Guidance:** Random dropout replaces 10% of the batch with learnable parameters [18].
- **Class-based Guidance (optional):** Random dropout replaces 10% of the batch with a null token.
- Jointly train conditional and unconditional models.
- At inference, we combine velocity predictions using two **scaled** extrapolations with:

w = 0 Unconditional model
w = 1 Standard conditional model
w > 1 Focuses sampling on high-probability areas (modes), by sharpening the distribution.

Qualitative Results

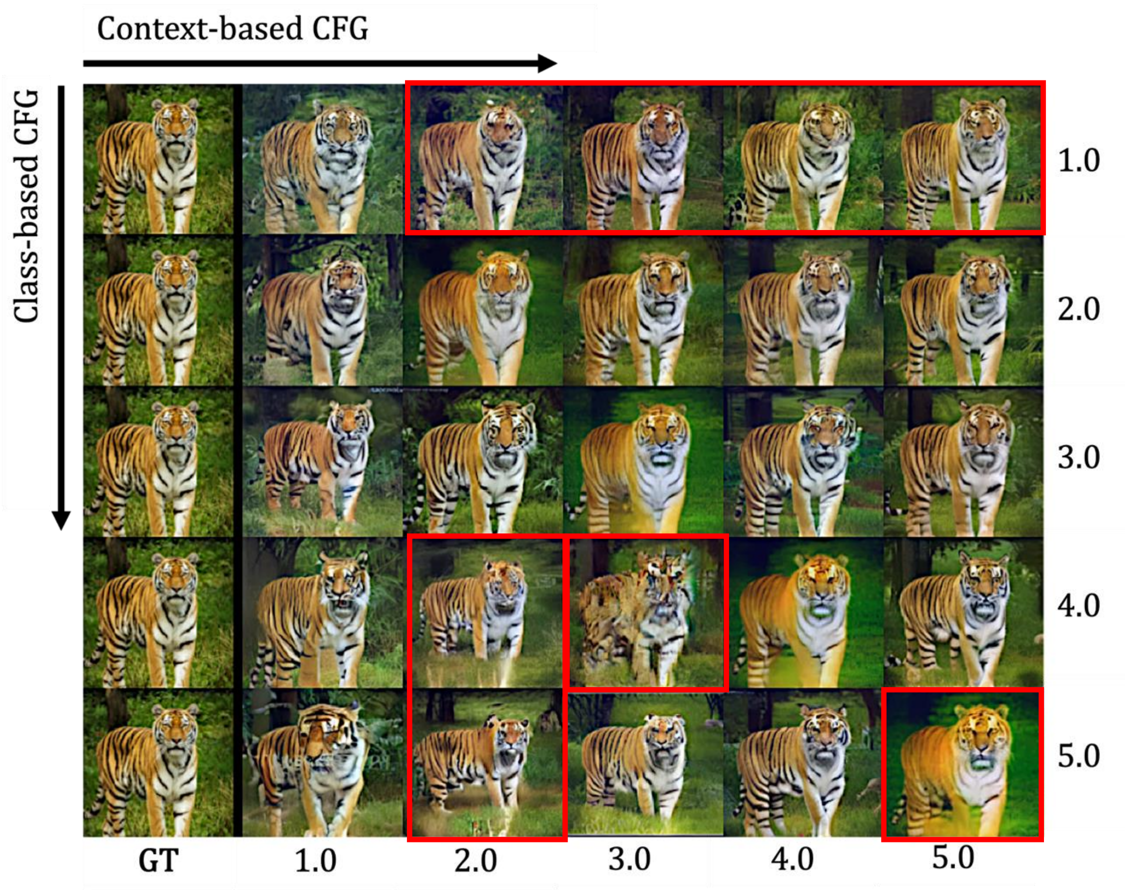
Evaluation of Custom Classifier-free Guidance Signal on Reconstruction Fidelity



- **Each reconstructed** sample is synthesized with custom classifier-free guidance-based algorithm (Self-Guidance = contextual features, Class-based Guidance = class label specificity).
    - **Self-Guidance**: outputs are contextually/structurally aligned (location, position, silhouette, colour, high-level consistency).

Qualitative Results

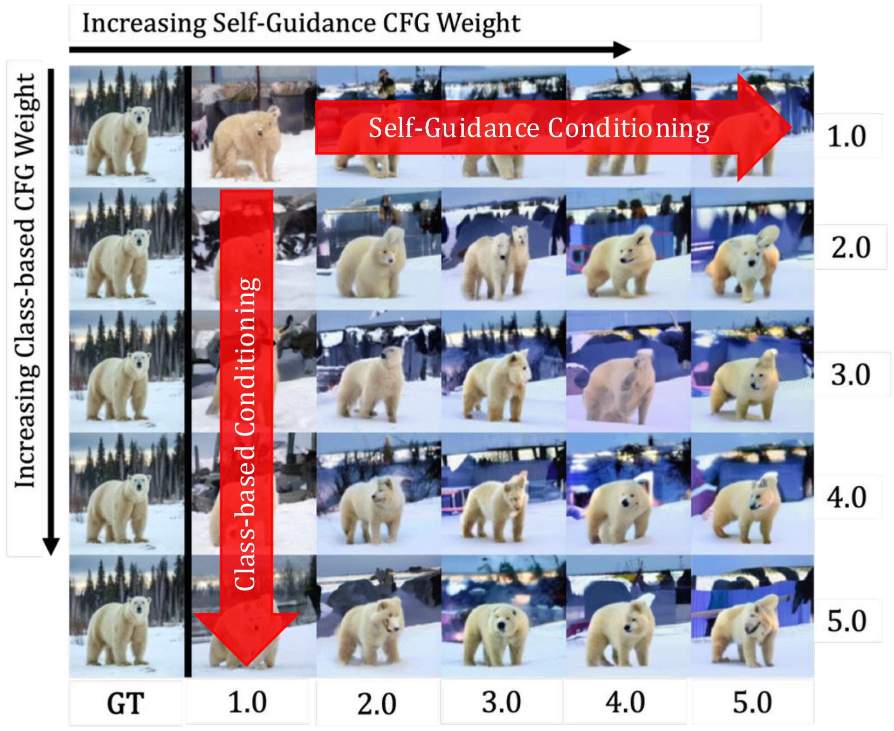
Evaluation of Custom Classifier-free Guidance Signal on Reconstruction Fidelity

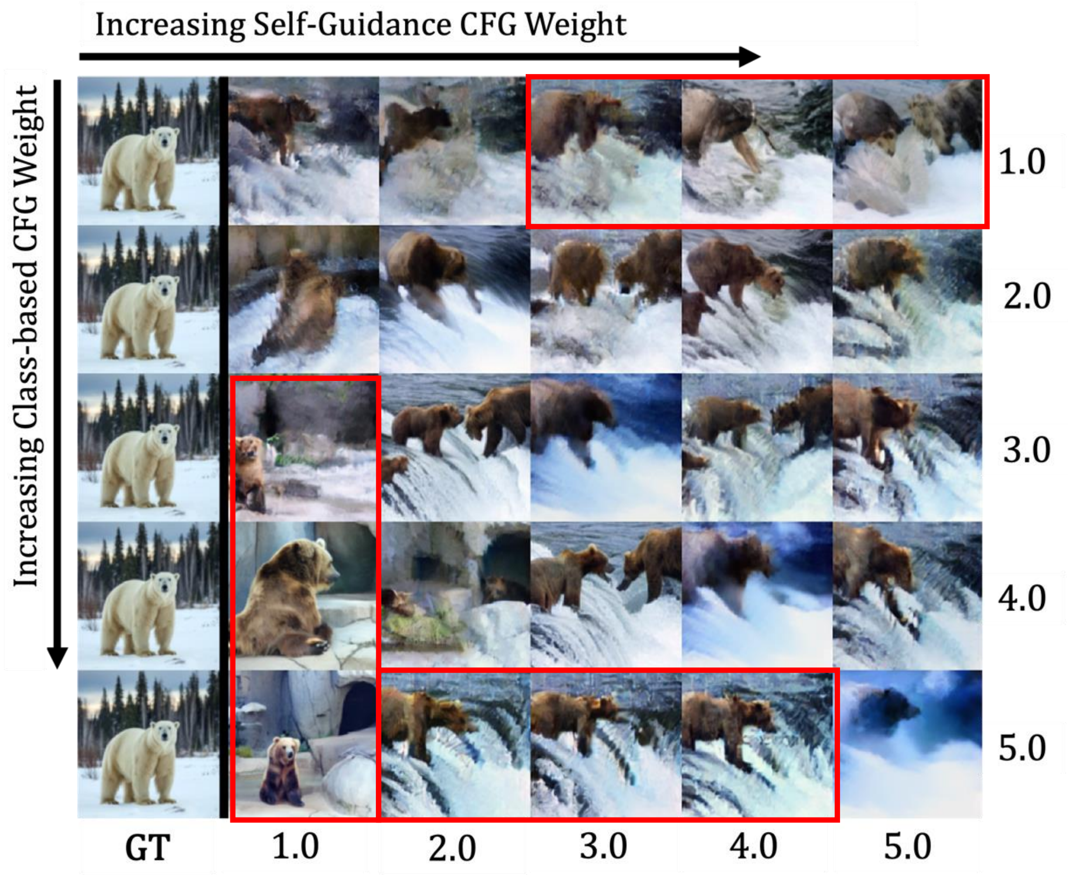


- **Each reconstructed** sample is synthesized with custom classifier-free guidance-based algorithm (Self-Guidance = contextual features, Class-based Guidance = class label specificity).
  - **Self-Guidance**: outputs are contextually/structurally aligned (location, position, silhouette, colour, high-level consistency).
  - **Class-Conditional**: outputs are less aligned with ground-truth and show larger variety (close-up, standing in water).
  - **Joint Conditioning:** balances both signals, combining high-level structure/context with class information.
- **Limitation:** Increasing joint conditioning scale reduces entropy, prunes fine-grained detail, and can lead to oversmoothed, oversaturated results, or even overshooting at very high scales.

Qualitative Results

Evaluation of Embedded Class Information in Self-Guidance z_t by Cross-Class Conditioning





Linear Interpolation

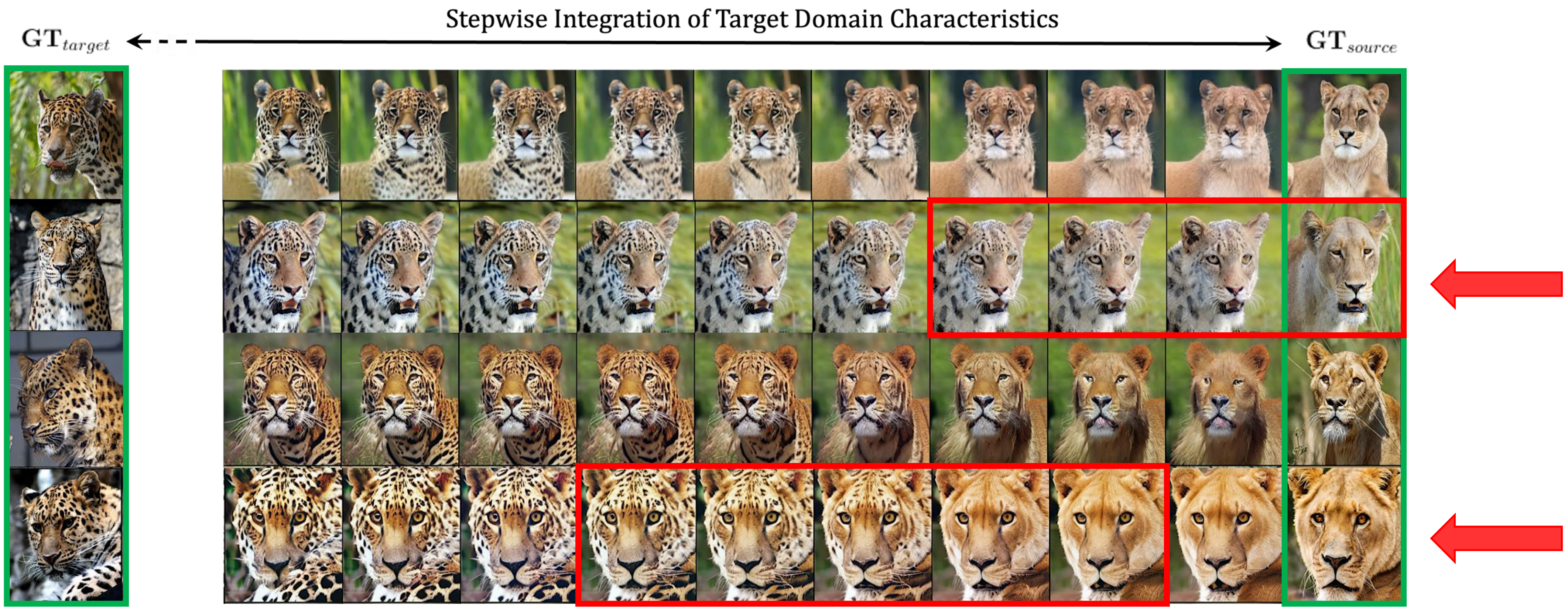


Figure: Manipulation of fur-patterns, while background, posture, facial expression of source remains the same. Allowing to assess learned concepts and closeness of specific colour combinations and patterns in Bayesian latent space.

Linear Interpolation

Evaluation of Learned Concepts and Bayesian Latent Space Smoothness

Red Panda → Giant Panda

Linear Interpolation

Evaluation of Learned Concepts and Bayesian Latent Space Smoothness

Snow Leopard → Tiger

“Sticky” patterns/textures:

- Reconstructed, high-frequency, fine-grained features such as texture and color patterns (e.g., tiger fur) may persist longer when interpolating, while low-frequency, coarse-grained features such as posture can evolve more rapidly, producing a perceptual “sticky” effect
- Linear traversal in β-VAE latent space may still produce **nonlinear pixel-level changes** due to the learned mapping of the self-conditioned DiT-XL/2 decoder to the image manifold
- β-VAE enforces *distributional regularization* for smooth latent representations, but latent axes are not semantically guaranteed. Hence, features aligned better with latent directions may change earlier than others.

Linear Interpolation

Evaluation of Learned Concepts and Bayesian Latent Space Smoothness

Siberian Husky → White Wolf

Linear Interpolation

Evaluation of Learned Concepts and Bayesian Latent Space Smoothness

Polar Bear → Brown Bear

Linear Interpolation

Across Domains

Conclusion and Recommendation

Conclusion

- Highlight the potential to extract **high-level semantic information** from noisy conditional probability paths through self-supervised representation learning.
- Utilising a self-supervised β-VAE encoder that defines a compact, auxiliary smooth latent space which enables both more **interpretable and decodable** latent manipulations.
- A **frequency-aware**, decoupled design separates semantic low-frequency forward processes from high-frequency reverse-time paths, allowing for more specialised functional roles.
- Representation learning-based **self-guidance** improves generative **steerability** and strengthens alignment with conditioning signals, allowing latent perturbations to be faithfully decoded into image space.

Conclusion and Recommendation

Limitations

- Achieving **disentanglement** in Bayesian latent space while maintaining **high reconstruction accuracy** with the ground-truth remains a challenging trade-off since fine-grained information and class-based information is increasingly lost – especially under high input noise levels.

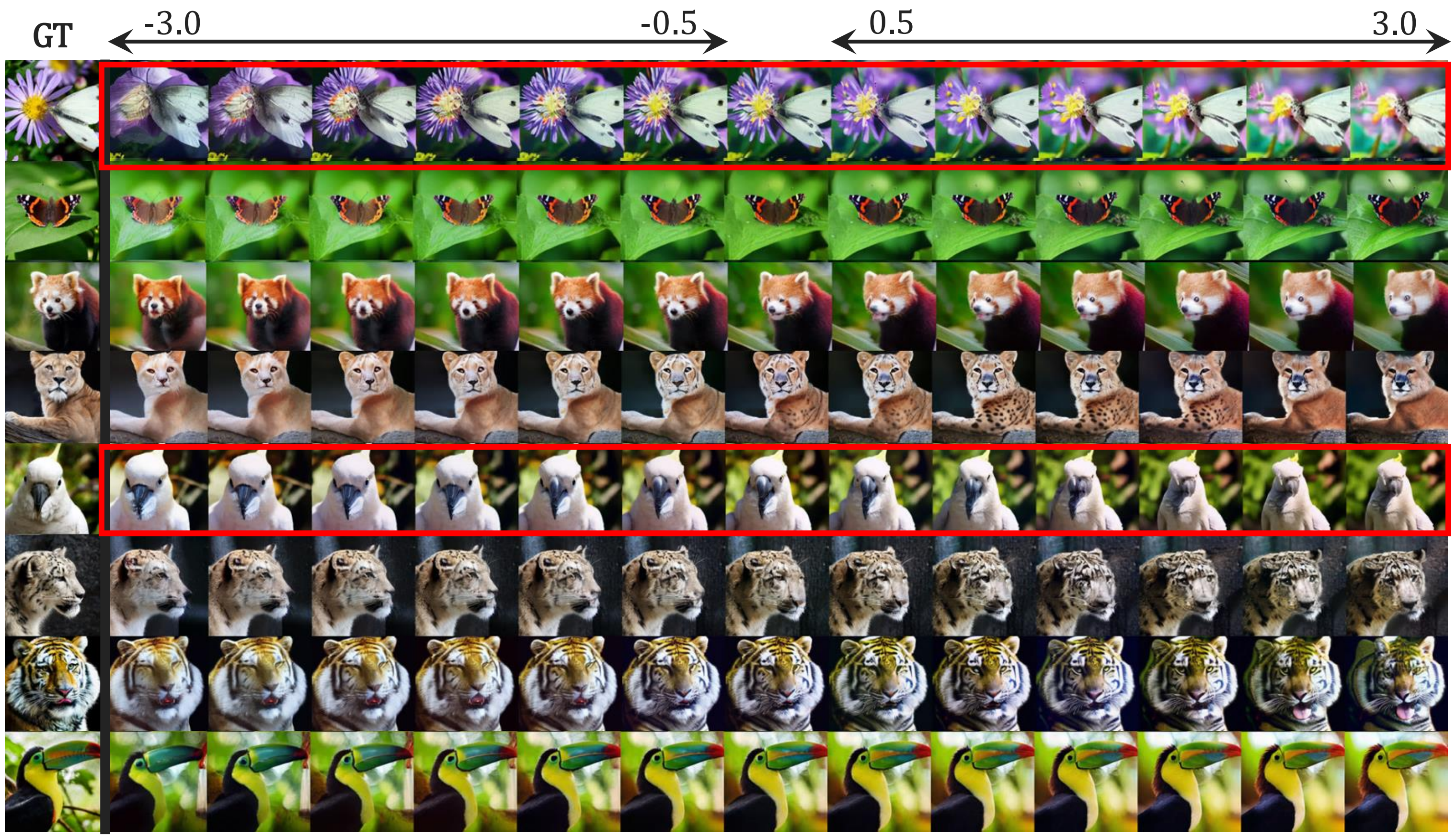
- Further development of **quantitative metrics** is needed to better assess **latent space quality and smoothness**, building upon existing measures such as **L-PPL** and **L-ISDT** [14-15].

- **Mechanistic interpretability** has limitations [23].
  While tools like **linear interpolation** (or latent walks) in learned ß-VAE latent space are scalable and valuable for building intuition for concepts learned and latent space quality through semantic probing, **more complex behaviours** could be non-linearly encoded, or harder to disentangle, limiting the effectiveness of **reverse-engineering approaches** for multi-factorial or abstract concepts.

- Requires additional human inspection, due to which it is more challenging to fully automate such qualitative, visual methods.

Future Directions

Manual Evaluation of Meaningful Directions in Self-Guidance z_t



Attribute Manipulation

- Another way to assess the relationship between image semantics and linear motion in the latent space is by moving the z_t in a particular **direction** to observe changes.
- **Manual search** for directions for attribute manipulation on real images can be unreliable and tedious.

Open direction:

- Finding meaningful directions through the learned weight vector of a **linear classifier** trained on latent codes of negative and positive images of a target attribute (e.g. zoom) [11].

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Future Directions

3D Feature conditioning to improve spatial realism:

- Extending current conditional input by 3D information (e.g. point clouds, depth maps) to improve physical structure and geometric consistency of reconstructed objects [24].

  - ➢ Can allow more accurate movement and pose estimation during interpolation.
  - ➢ Better generalization in unseen poses.
  - ➢ Potential applications in creative domains

Temporal extraction and fusion to improve continuity:

- Capture not only high-level semantic information (e.g. colour distribution) but also **temporal information** in a separate vector from noisy latent codes and how images evolve over time [25].

  - ➢ Might better capture how images evolve over time not only static features at a specific timesteps.
  - ➢ Could enhance temporal coherence in long-range interpolation.

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Appendix

Appendix: Literature Review

Literature Review Findings

"A Noise is Worth Diffusion Guidance" (Ahn et al., 2024)

- Certain noise patterns obtained via **diffusion inversion** can reconstruct high-quality images **without guidance**.

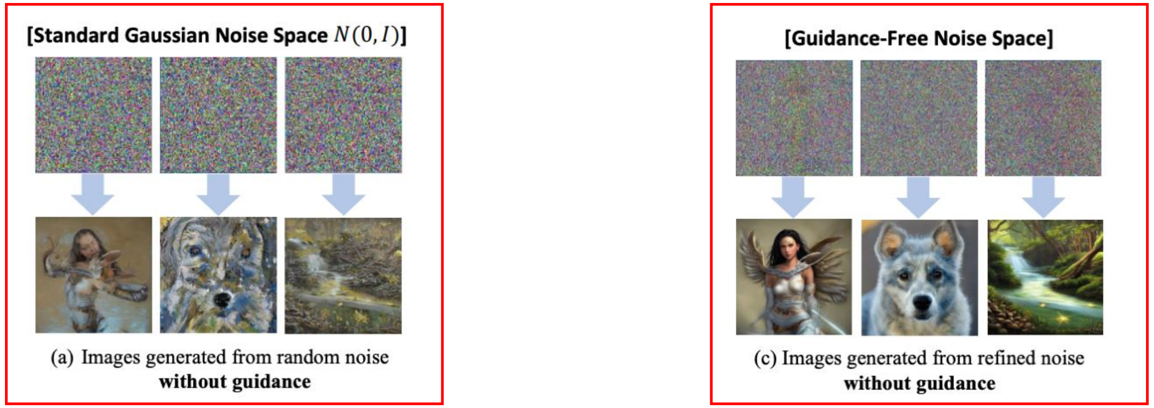


Figure: **Difference between initial noise.** Diffusion models often fail to generate high-quality images without guidance. However, guidance-free noise spaces can achieve this.

[5] Ahn, D., Kang, J., Lee, S., Min, J., Kim, M., Jang, W., et al. (2024). A Noise is Worth Diffusion Guidance. arXiv preprint arXiv:2412.03895.

Literature Review Findings

"A Noise is Worth Diffusion Guidance" (Ahn et al., 2024)

Guidance-Free Noise Space

- First denoise an initial noise $\boldsymbol{x}_T$ to clean image $\boldsymbol{x}_0^{Guide}$ using a text prompt with classifier-free guidance
- Invert image to create noise that can reconstruct the same image without guidance.

$$x_T^{\text{Guide}} := \text{Inversion}(\text{Denoise}_{\text{Guide}}(x_T, c))$$

- Find mainly **low-frequency** semantic information drives reconstruction quality.

➡ DDIM inversion leads to artefacts

Training Methodology

- **NoiseRefine model**, a small ResNet with residual connections and LoRA, to learn a mapping from Gaussian noise to a guidance-free noise.
- Shift to **pixel-space loss** rather than noise-space loss to avoid inversion errors.
  - ➢ Noise refining model estimates the refined noise $\hat{\boldsymbol{x}}_T = g_\phi(\boldsymbol{x}_T)$
  - ➢ Pass noise to diffusion pipeline which generates an image without guidance.
  - ➢ Optimise distance loss to reduce gap between clean image and reconstruction.

$$d(\hat{x}_0, x_0^{\text{Guide}}) = \left\|\hat{x}_0 - x_0^{\text{Guide}}\right\|_2^2$$

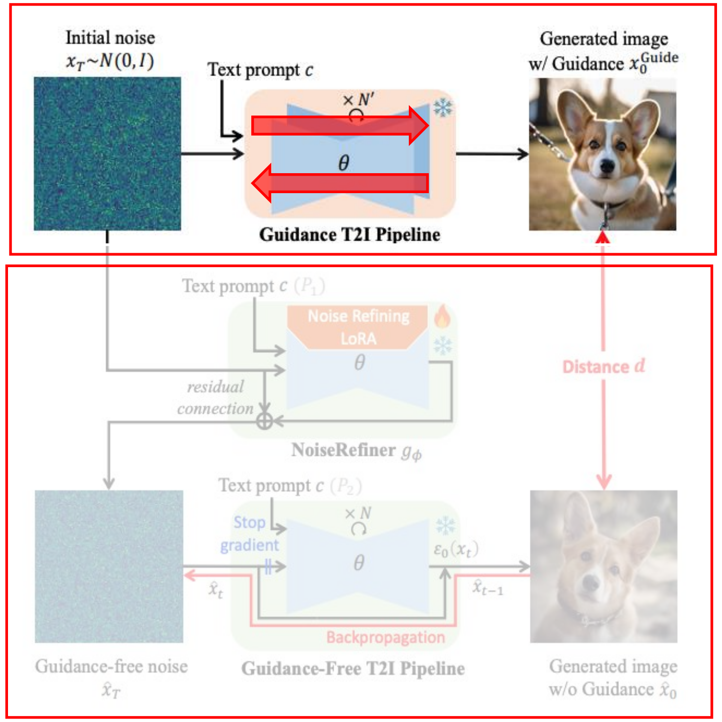


Figure: **Training pipeline.** Learn a mapping from initial Gaussian noise to guidance-free noise samples.

[5] Ahn, D., Kang, J., Lee, S., Min, J., Kim, M., Jang, W., et al. (2024). A Noise is Worth Diffusion Guidance. arXiv preprint arXiv:2412.03895.

Literature Review Findings

"DDT: Decoupled Diffusion Transformer" (Wang et al., 2025)

- Builds on linear-interpolant-based Flow Matching with Diffusion Transformer
- Frequency components evolve along path between noise and data
  - **Forward noising:** preserve low-frequency structure
  - **Reverse denoising:** reconstruct high-frequency detail
- **Identifies an optimization dilemma** in coupled diffusion-transformers – encoding low-frequency semantics suppresses high-frequency details, creating tension between semantic extraction and fine-grained decoding

Key Contributions:

- **Decouples both stages** of the diffusion forward and reverse process.
- Introduces **Self-Conditioning** via a conditional Diffusion Transformer encoder, aligned through REPAlign with DINOv2 visual representations.

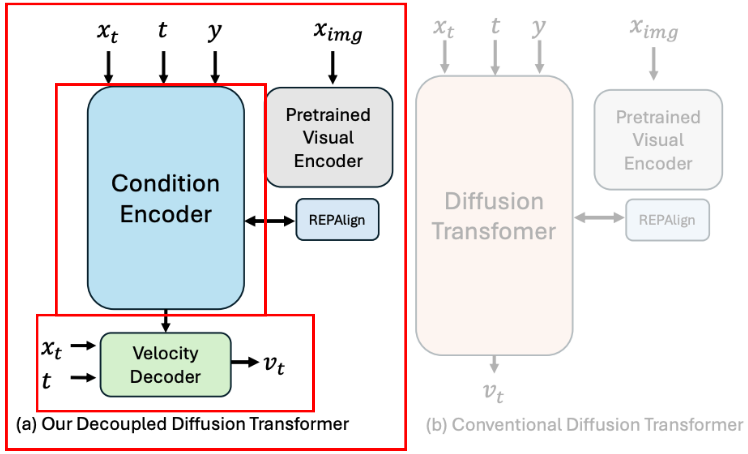


Figure: **Training pipeline.** Decoupled Diffusion Transformer incorporating a conditional encoder extracting semantic self-conditioning and a self-conditioned velocity decoder.

[6] S. Wang, Z. Tian, W. Huang, and L. Wang, "DDT: Decoupled Diffusion Transformer," arXiv preprint arXiv:2504.05741, 2025.

Literature Review Findings

”Diffusion Autoencoders: Toward a Meaningful and Decodable Representation" (Preechakul et al., 2021)

- DiffAE explores the possibility to use diffusion for representation learning through an auxiliary, learnable encoder component
- The input image is encoded into a two-part latent code:
  - ➢ A **high-level, non-spatial semantic vector** z_{sem} capturing global information
  - ➢ A **low-level stochastic component** x_T derived via DDIM inversion for fine-grained details

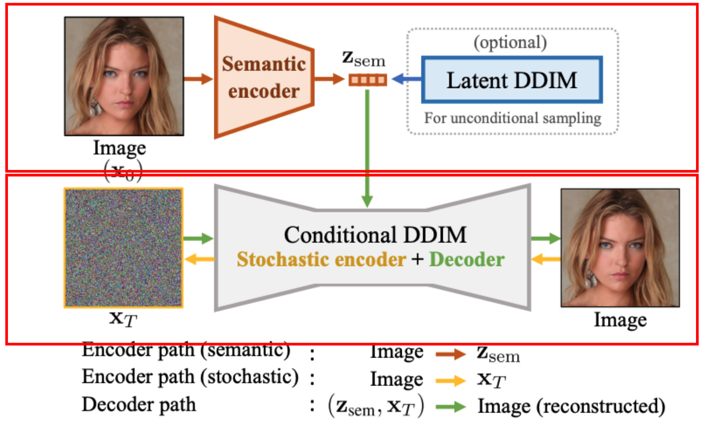


Figure: **Training pipeline.** The autoencoder consists of a semantic encoder that maps input to a semantic subcode and a conditional DDIM that defines a stochastic encoder and decoder.

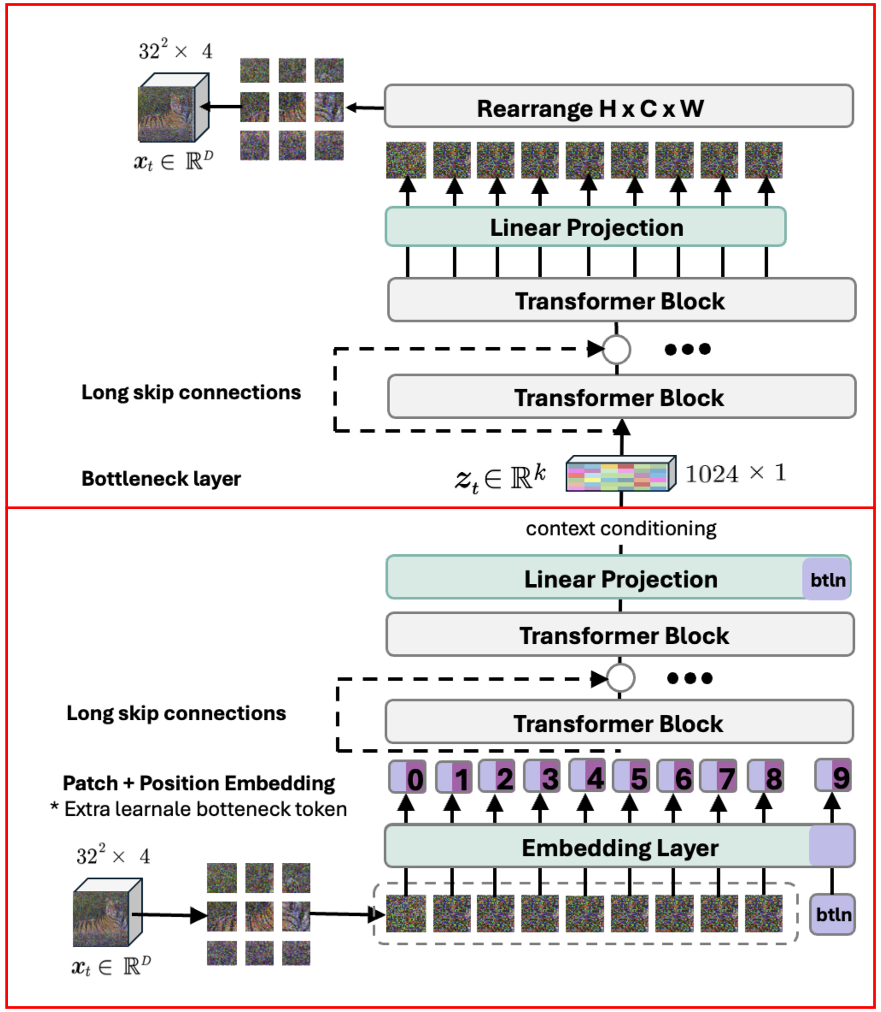
Key Contributions:

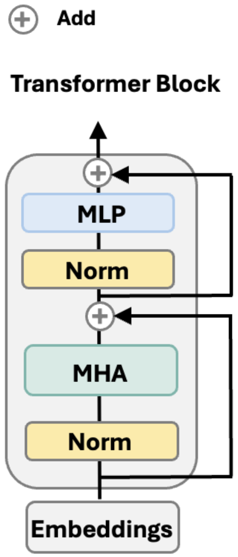
- Introduce a decodable, meaningful latent space for Diffusion Models.
- Uses DDIM as both decoder and stochastic encoder to generate latent noises .
- Enable interpolation and real-image attribute-level manipulation (e.g., female to male).

[10] K. Preechakul, N. Chatthee, S. Wizadwongsa, and S. Suwajanakorn, “Diffusion autoencoders: Toward a meaningful and decodable representation,” arXiv preprint arXiv:2111.15640, 2022.

Appendix: Methods

Beta-VAE Architectural Details



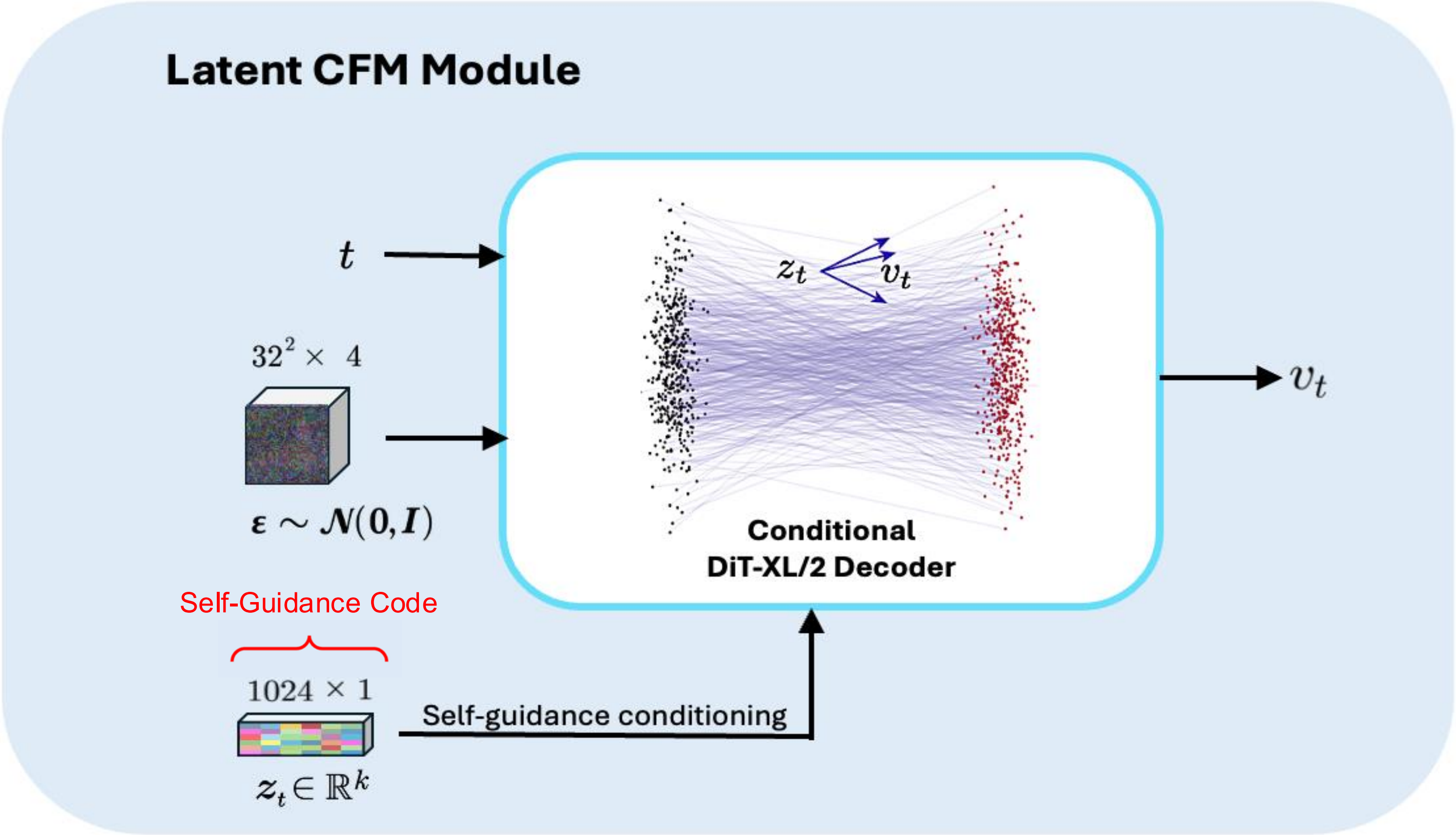


Architectural Design

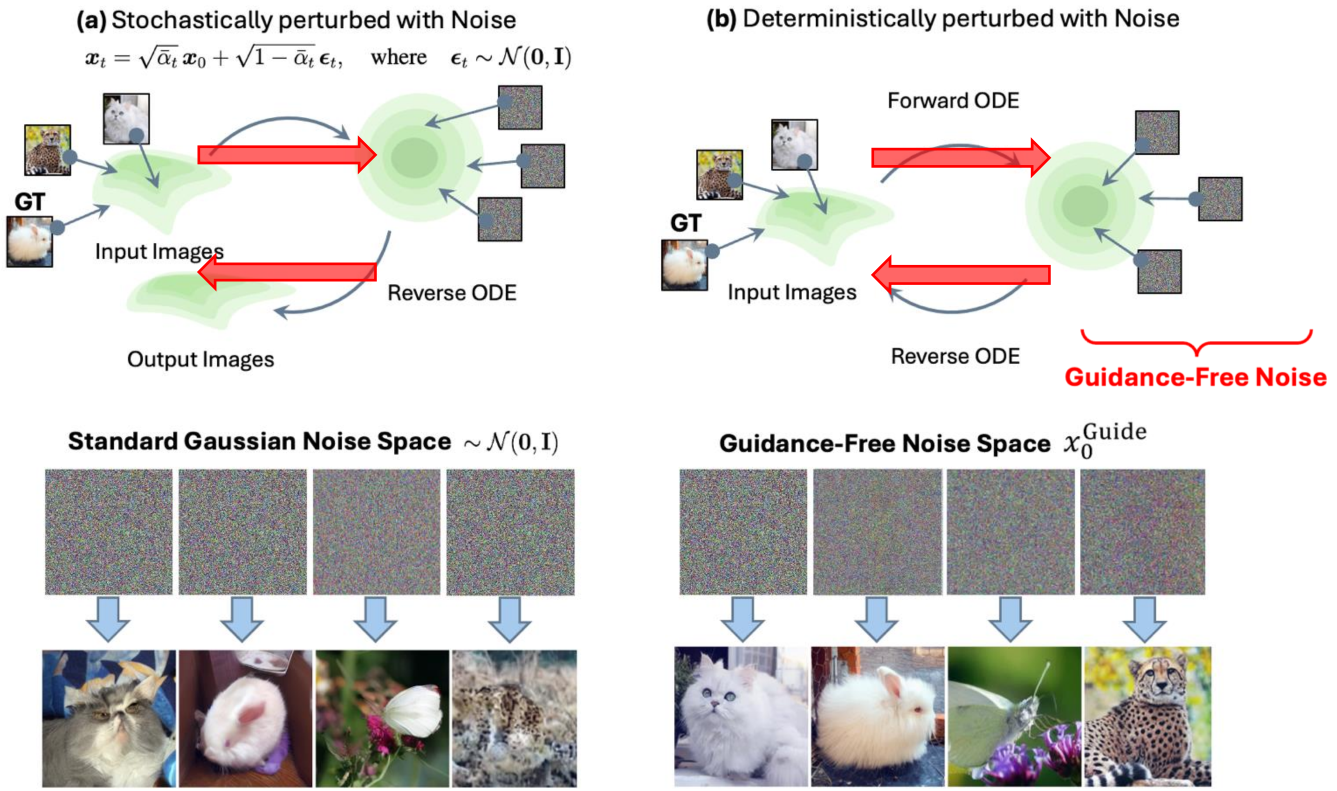
- Based on a standard ViT design with patch embedding and Transformer blocks.
- Introduces **cross-attention skip connections** in encoder and decoder to control the flow of information from earlier to later layers.
    - ➢ Improves **gradient flow** during training.
    - ➢ Preserves **fine-grained input details**, especially under high input noise settings.

Representation Learning Framework

Stage 2: Reverse-time High-Frequency Decoding



Guidance-Free Noise Space



(a) Images generated from samples with applied random noise - **without guidance.**

(b) Images generated from samples with applied deterministic noise - **without guidance.**

Guidance-free Noise

Under our optimal transport-based Flow Matching approach, the forward ODE noised images are encoded into unique initial noise samples are **"guidance-free"** [1-3, 5].

- Guidance-free: doesn't require additional guidance to generate high-quality image samples.

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Evaluation Metrics

Latent Perceptual Path Length (L-PPL)

$$\text{L-PPL} = \mathbb{E}_{\boldsymbol{z}_t^{(1)}, \boldsymbol{z}_t^{(2)} \sim \mathcal{P}(\mathcal{Z}),\ \lambda \sim \mathcal{U}(0,1)} \left[\frac{1}{\delta^2} \underbrace{d}_{\text{E-Latent LPIPS}}\big(G(\boldsymbol{z}_t^{(\lambda)}), G(\boldsymbol{z}_t^{(\lambda+\delta)})\big) \right],$$

1) Linearly interpolate: $\boldsymbol{z}_t^{(\lambda)} = \text{lerp}(\boldsymbol{z}_t^{(1)}, \boldsymbol{z}_t^{(2)}, \lambda)$
2) Generate sub-sequence over two close points: $G(\boldsymbol{z}_t^{(\lambda)}), G(\boldsymbol{z}_t^{(\lambda+\delta)})$, where δ is an infinitesimal small step.
3) Measure perceptual distance between the two latent points.
4) Average over path segments, normalized by the step size.

Evaluation of Latent Smoothness

- Well-behaved Bayesian space should show **smooth, gradual changes** (i.e., small change in latent space, results in small, meaningful change in output space).
- Abrupt changes may indicate **poorly structured** or “entangled" latent space.

(a) Latent Perceptual Path Length (L-PPL)

- Based on the Perceptual Path Length (PPL) metric introduced in StyleGAN [17] to assess **smoothness** or interpretability of the latent space.
- Measures how much the latent output **changes perceptually** (i.e., using perceptual E-Latent LPIPS distance) for a small step taken along the linear trajectory in Bayesian latent space.

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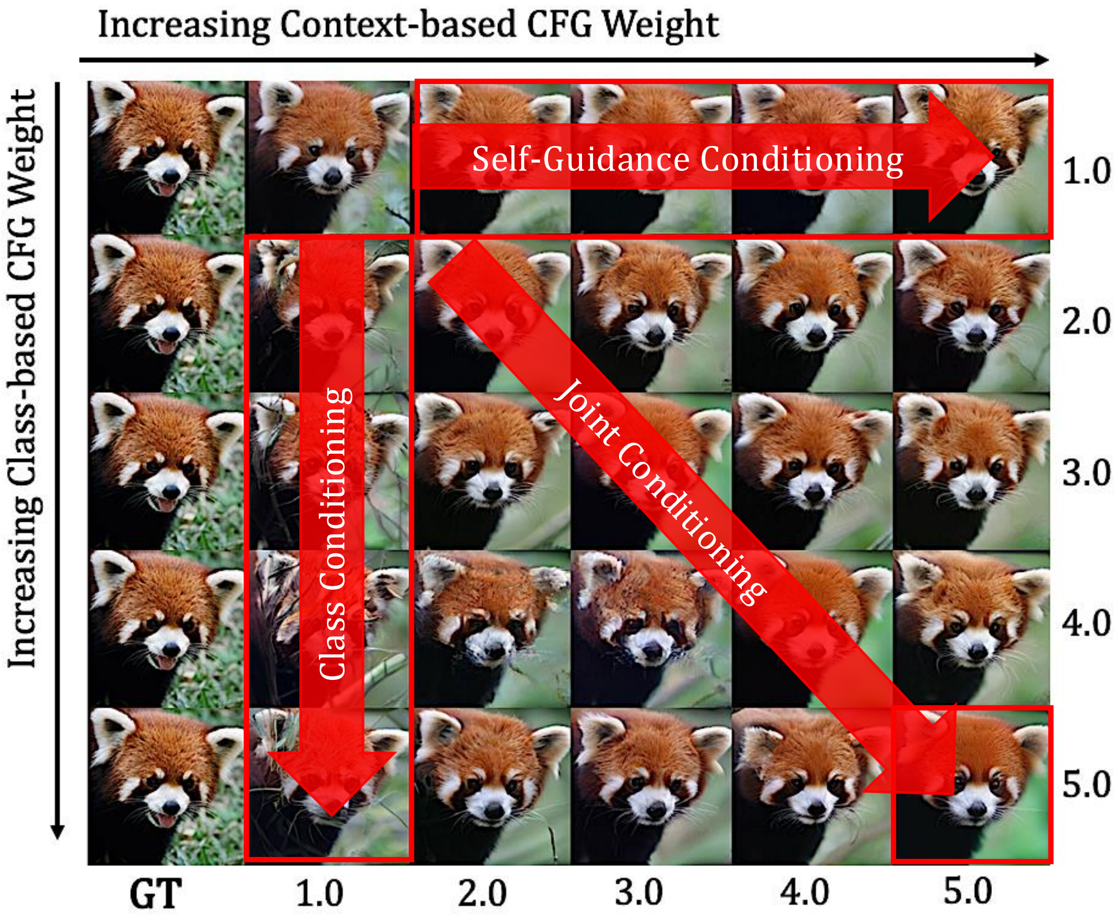
Appendix: Qualitative Results

Qualitative Results

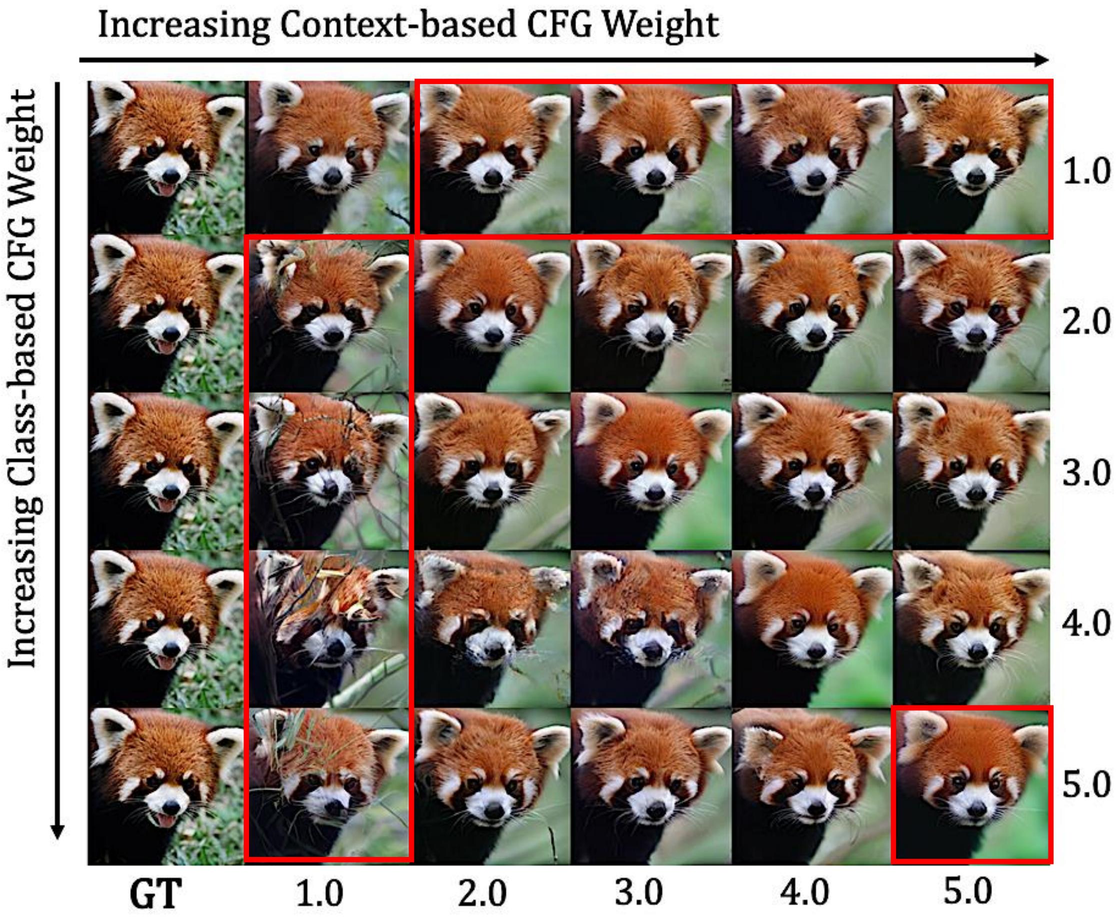
Evaluation of Custom Classifier-free Guidance Signal on Reconstruction Fidelity



- **Each reconstructed** sample is synthesized with custom classifier-free guidance-based algorithm (Self-Guidance = contextual features, Class-based Guidance = class label specificity).
    - **Self-Guidance**: outputs become contextually/structurally aligned (location, position, silhouette, colour, high-level consistency).
    - **Class-Conditional**: outputs are less aligned with ground-truth and show larger variety (close-up, standing in water).
    - **Joint Conditioning:** balances both signals, combining high-level structure/context with class information.
- **Limitation:** Increasing joint conditioning scale reduces entropy, prunes fine-grained detail, and can lead to oversmoothed, oversaturated results, or even overshooting at very high scales.

Qualitative Results

Evaluation of Custom Classifier-free Guidance Signal on Reconstruction Fidelity



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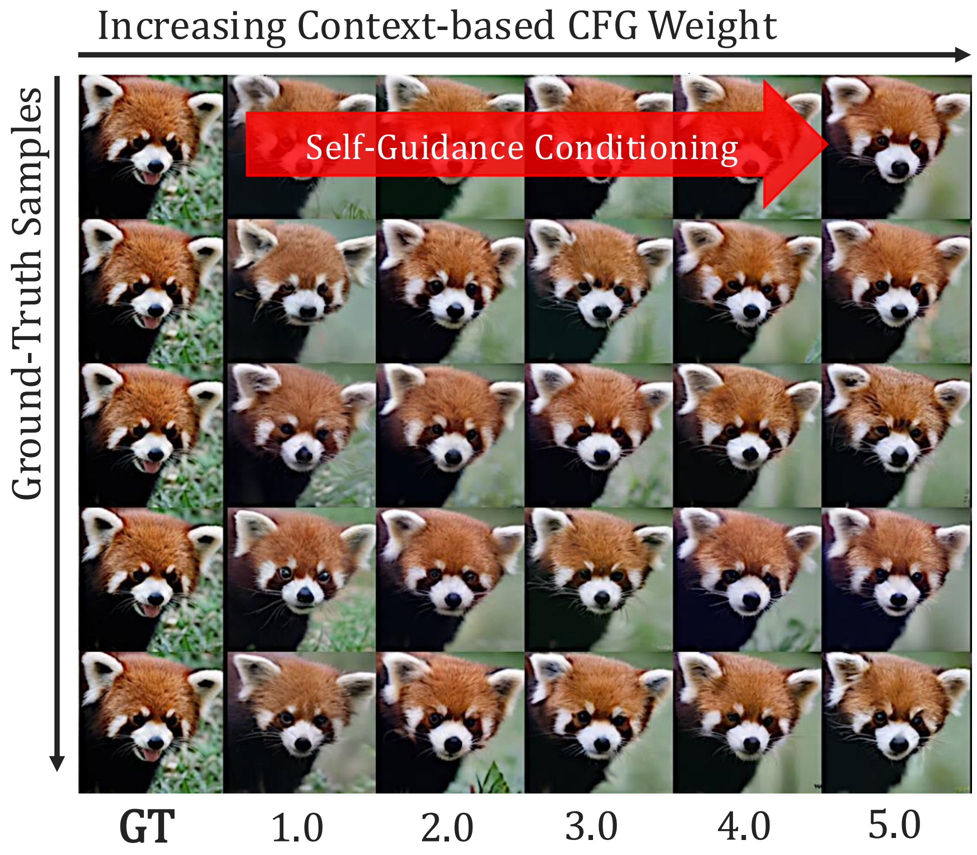
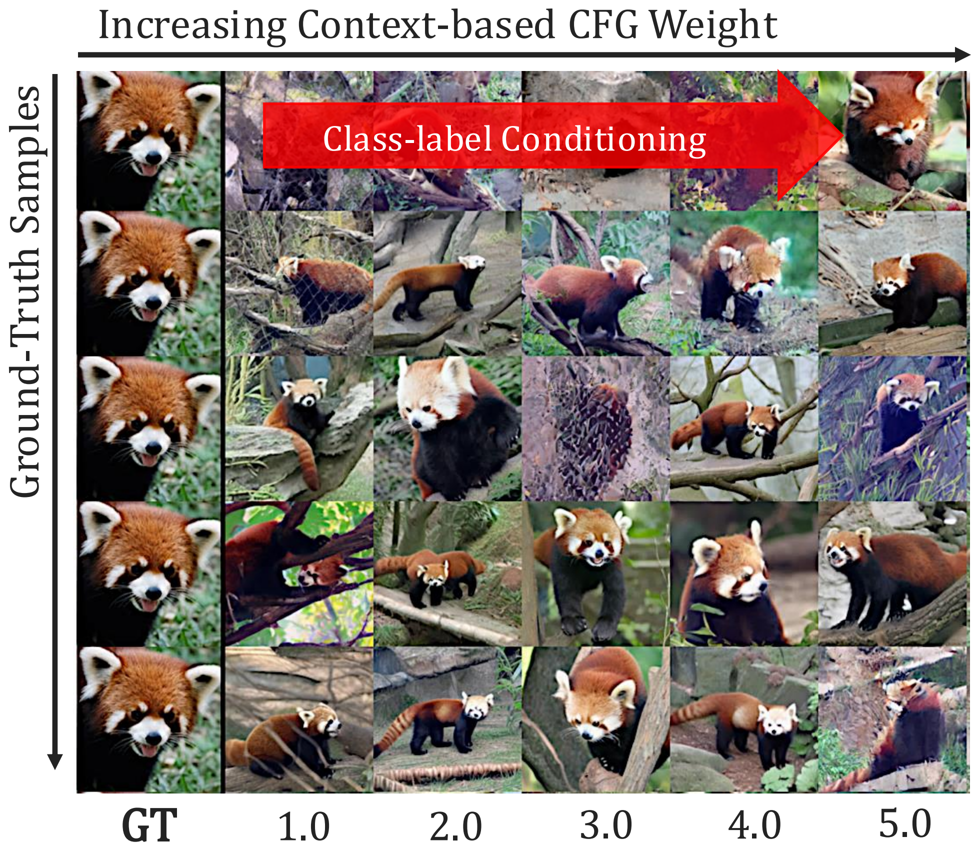
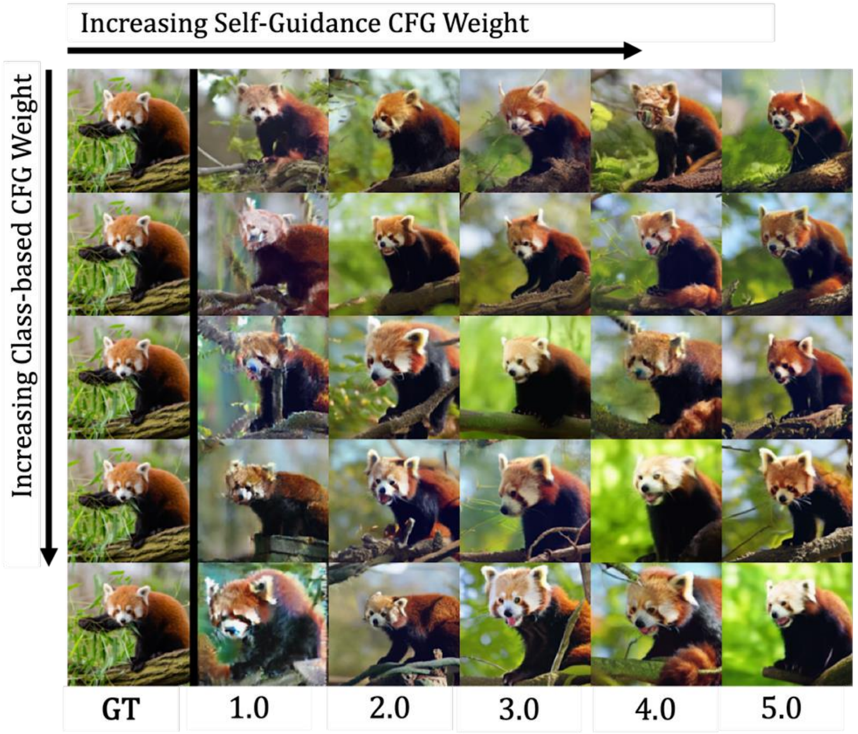
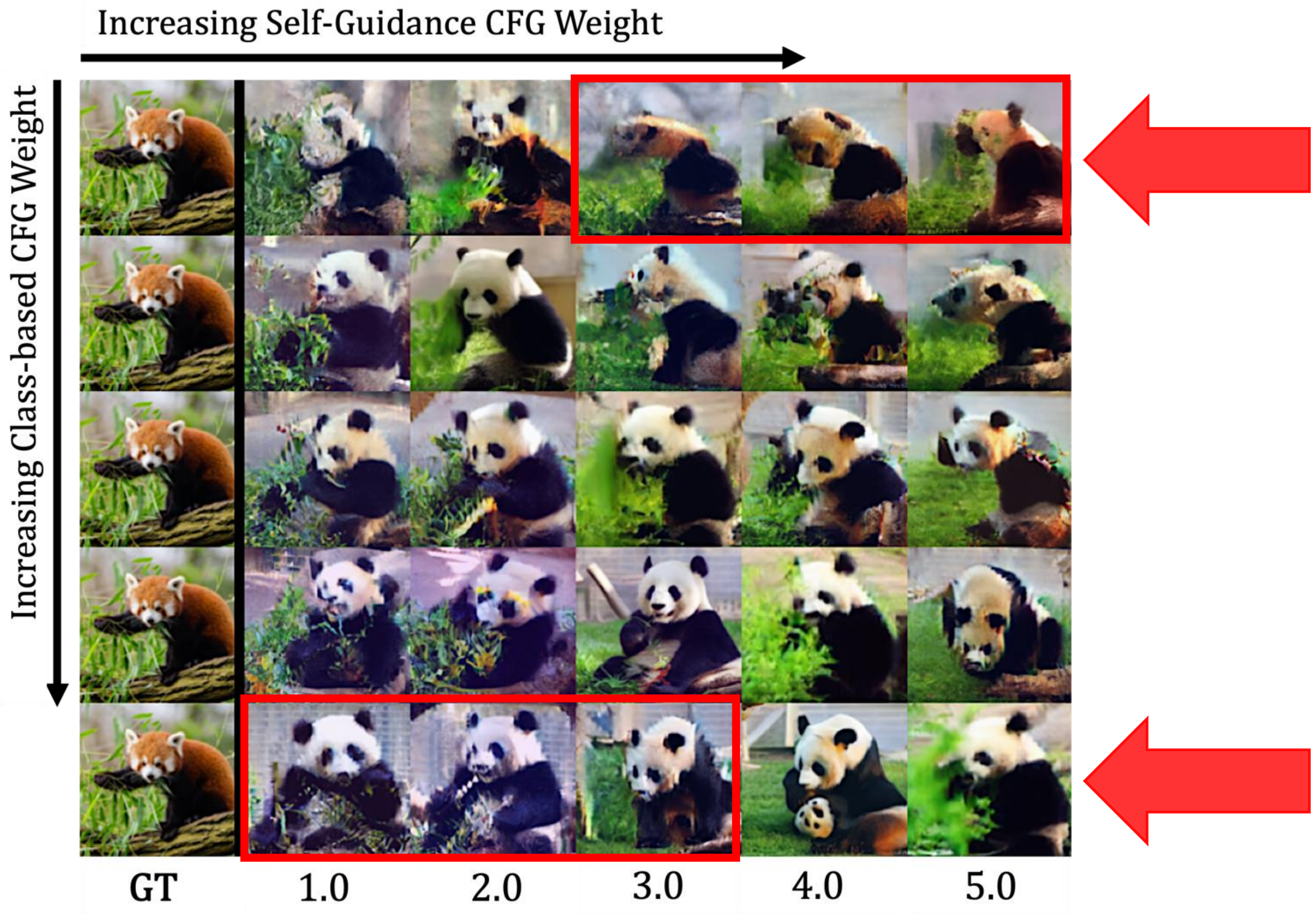




Qualitative Results

Evaluation of Embedded Class Information in Self-Guidance z_t by Cross-Class Conditioning





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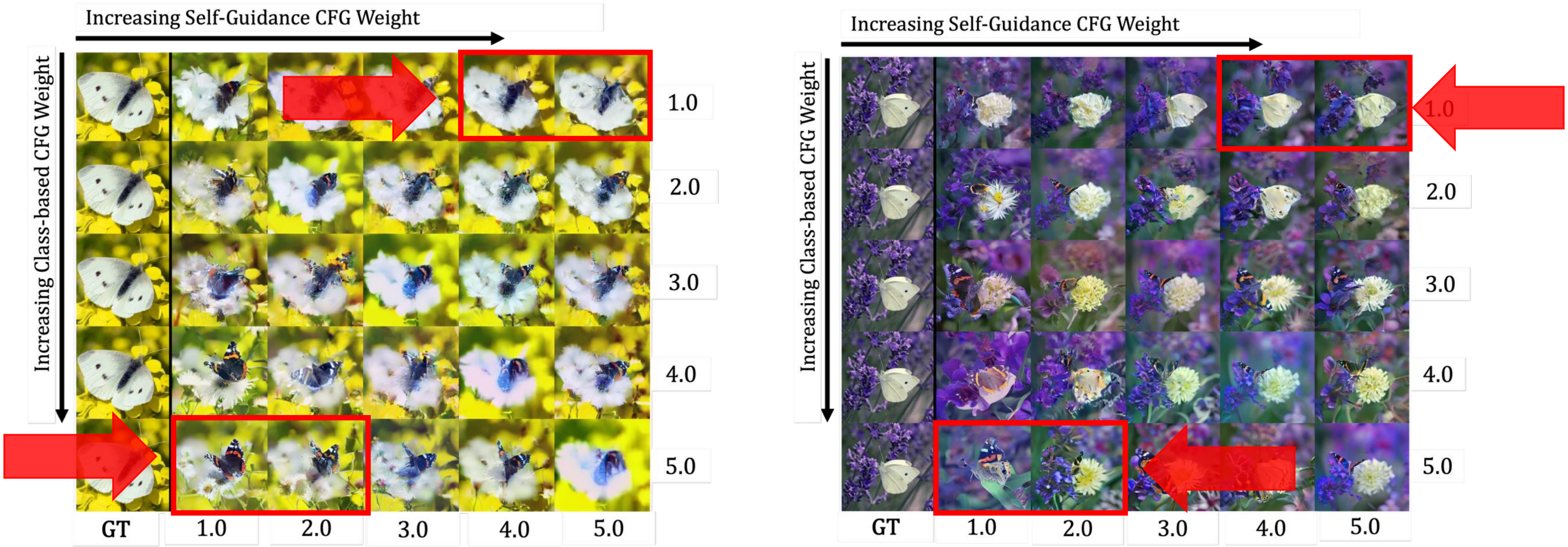


Figure: Learned concepts and class-based information. Left: white cabbage butterfly becomes flower background, right: white cabbage butterfly becomes flower background with increasing class-based conditioning, whereas it remains in butterfly shape under increased self-guidance weight.

Qualitative Results

Evaluation of Embedded Class Information in Self-Guidance z_t by Cross-Class Conditioning

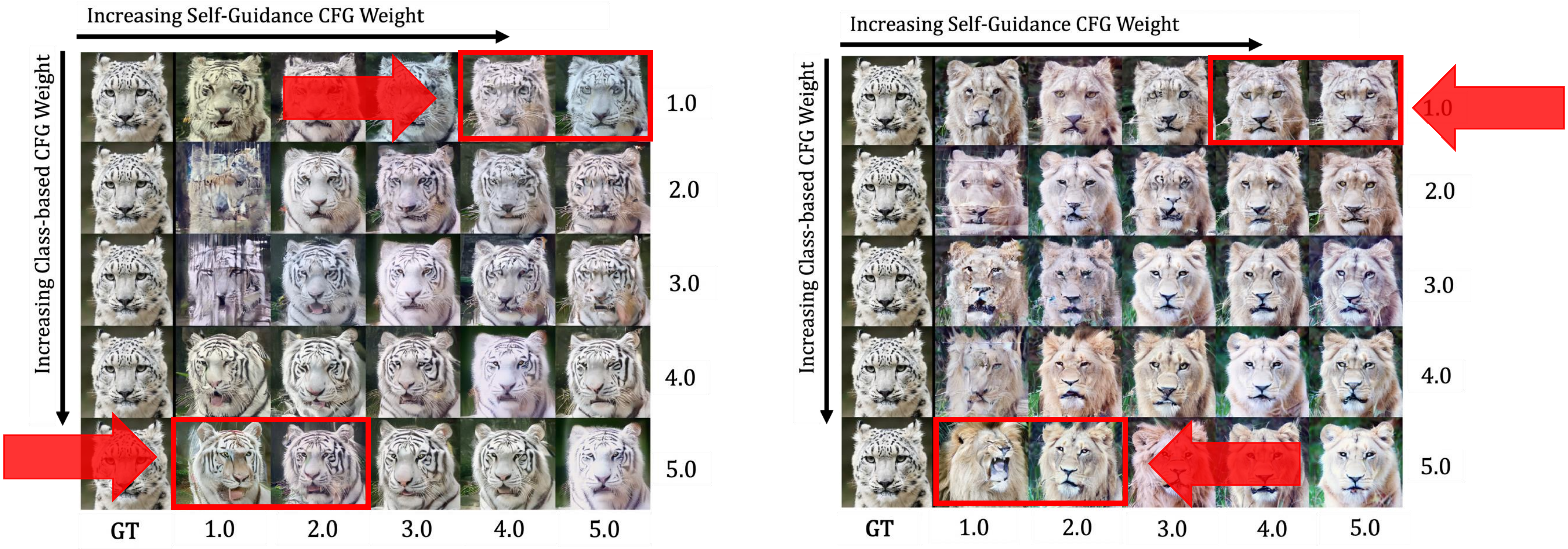
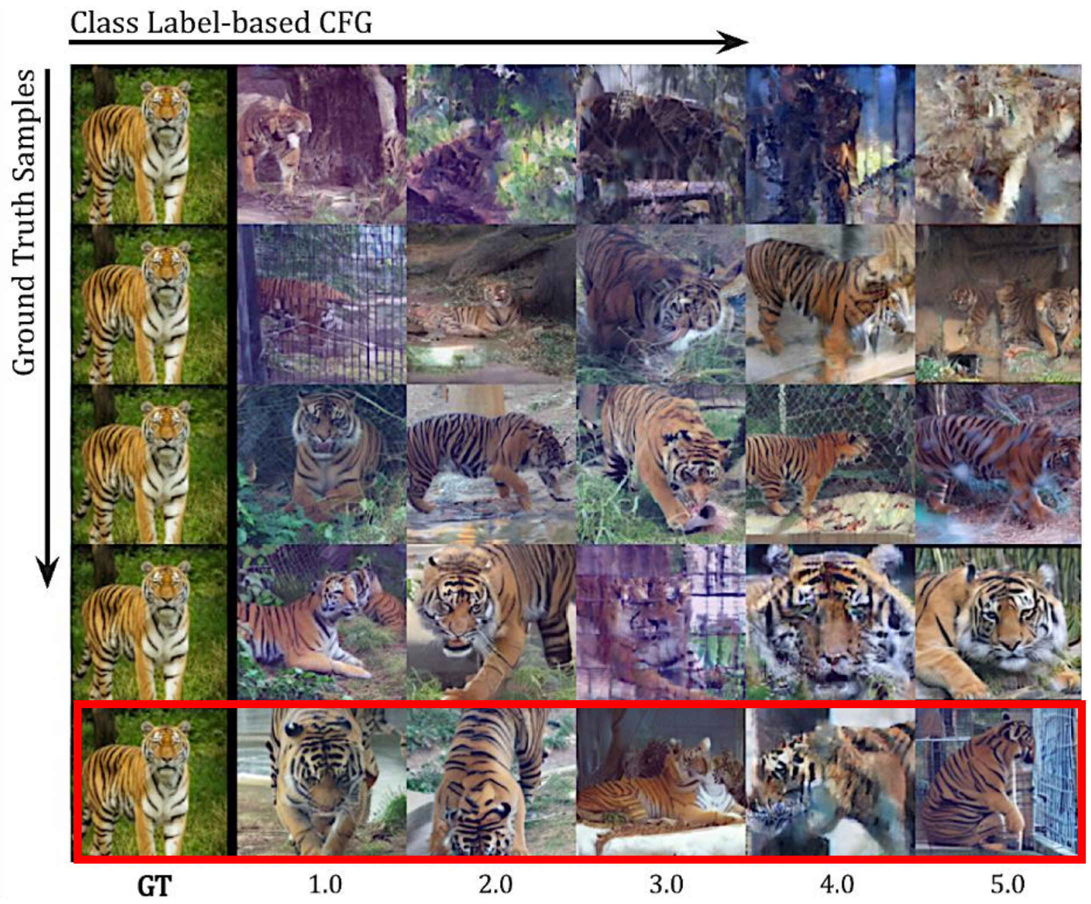


Figure: Learned concepts and class-based information. Left: Add patterns, right: Remove patterns, whilst maintaining posture, fur colour, facial expression.

Qualitative Results

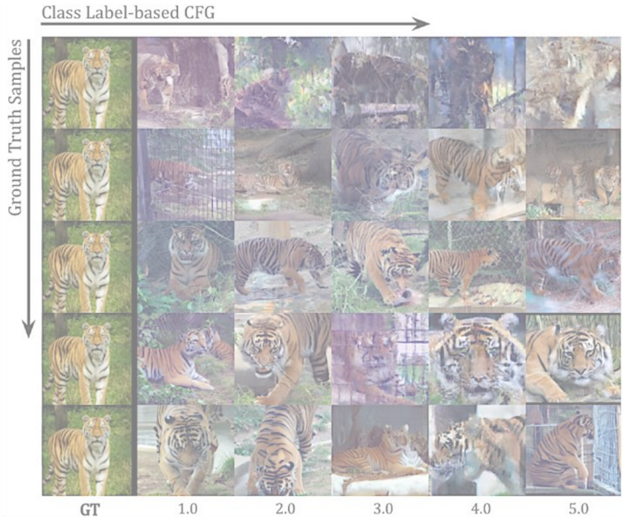
Evaluation of Custom Classifier-free Guidance Signal on Reconstruction Fidelity



- **Each reconstructed** sample is synthesized with classifier-free guidance utilising only class labels with varying initial noise samples.
  - ➢ First column: ground truth samples for comparison.
  - ➢ Rows: from left to right increasing CFG scales.
- The outputs exhibit greater "randomness" with respect to the original input, highlighting samples most likely under the learned class-conditional data distribution but poorly aligned with the original input.

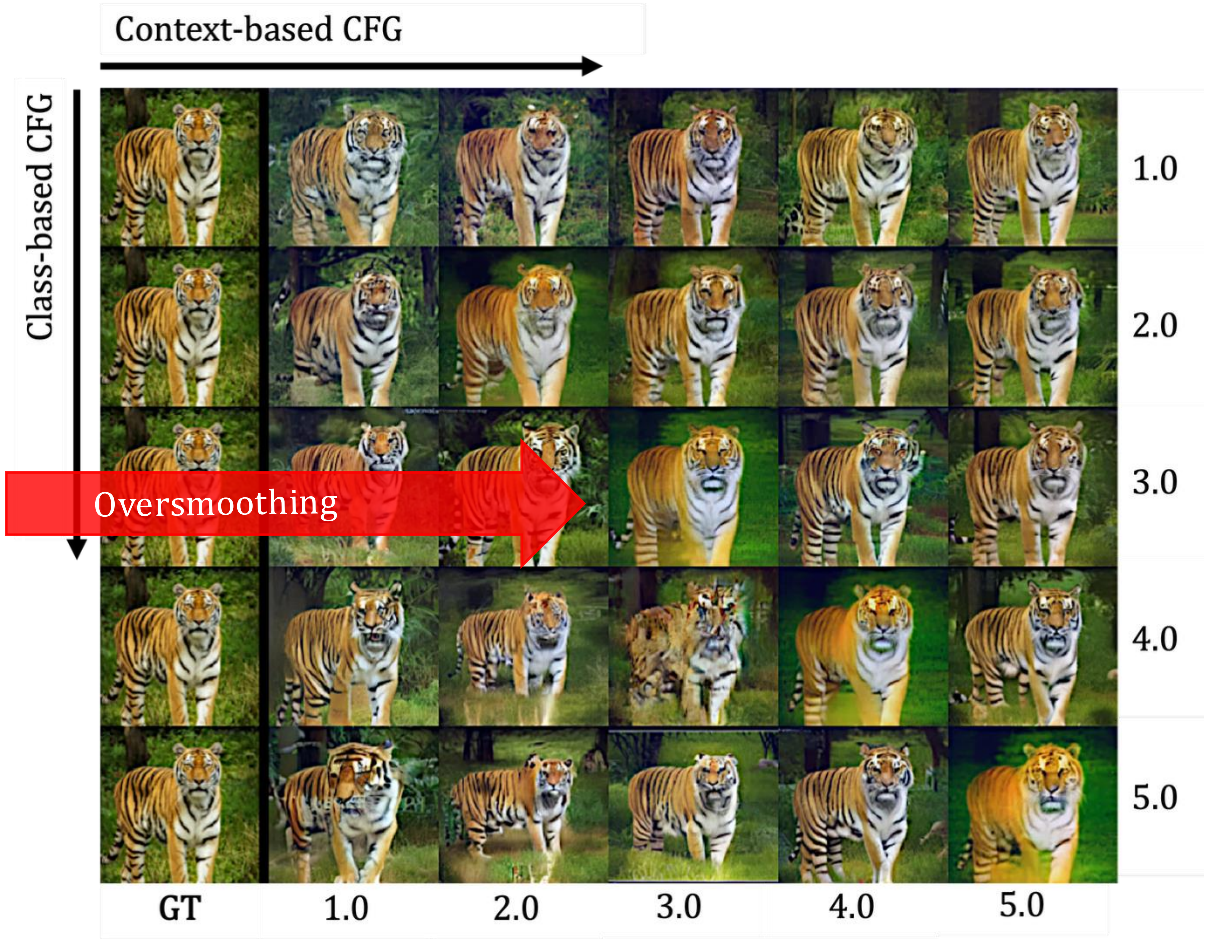
Qualitative Results

Evaluation of Custom Classifier-free Guidance Signal on Reconstruction Fidelity

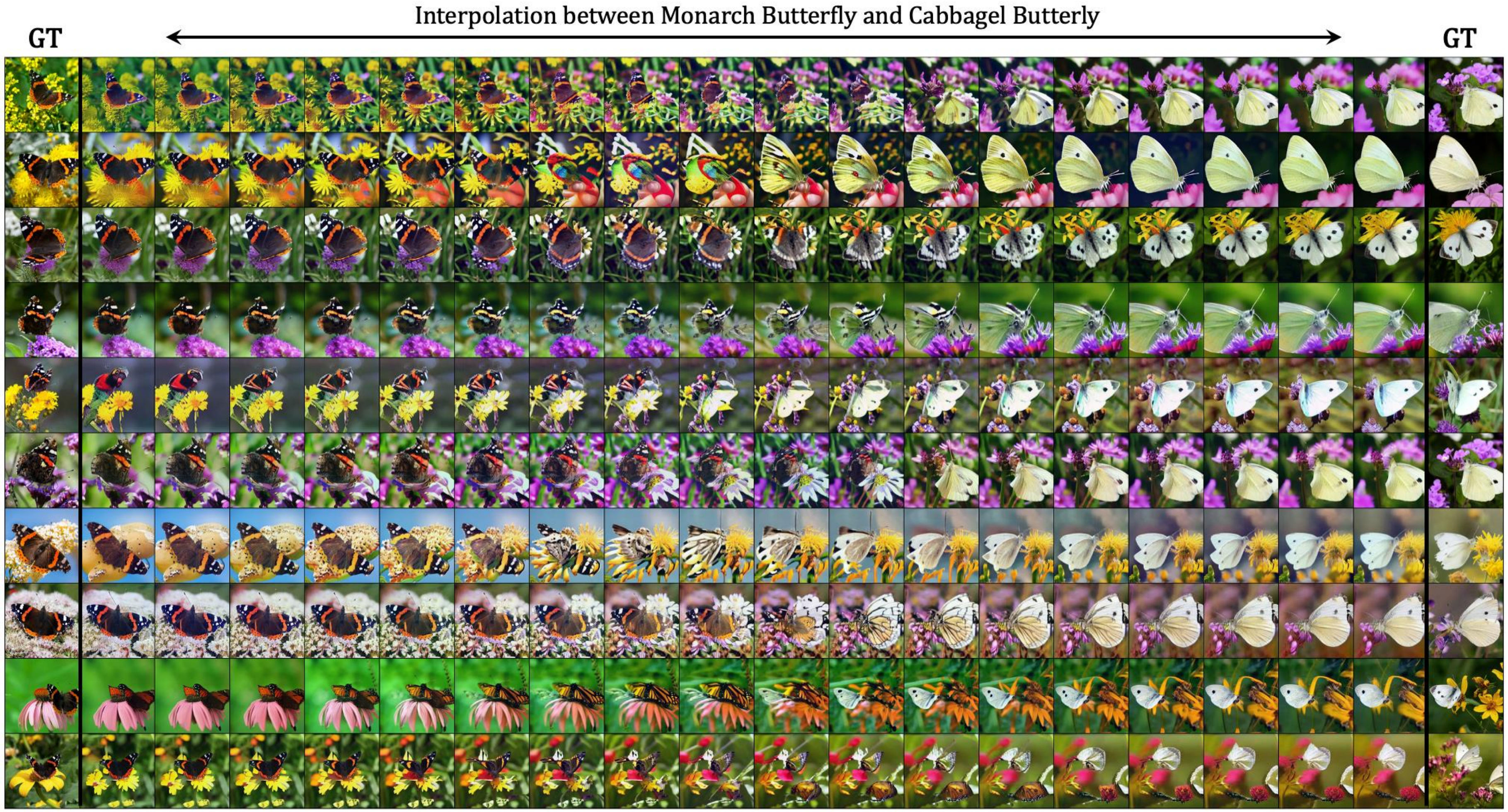


Qualitative Results

Class Label- and Self-Guidance/Context-based Classifier-Free Guidance Matrix

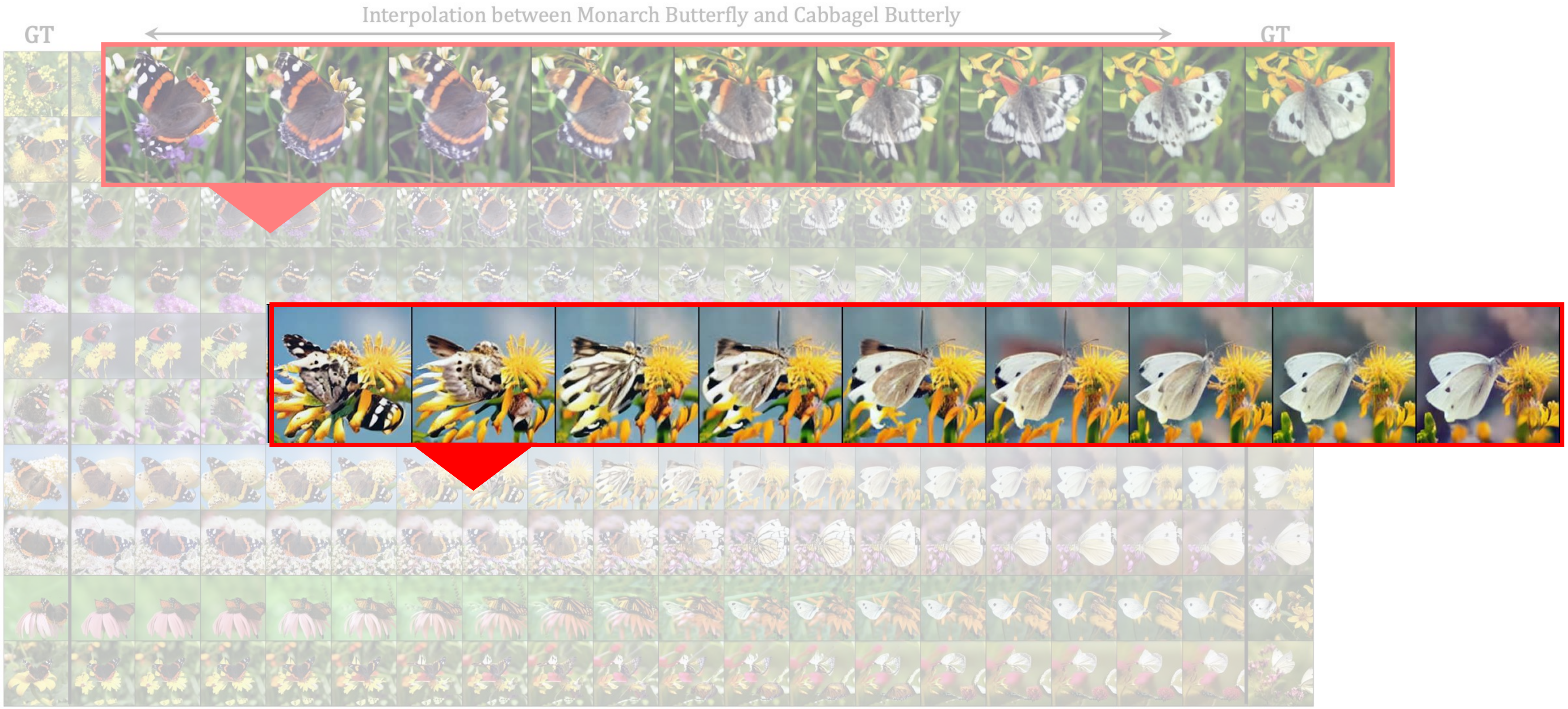


Linear Interpolation



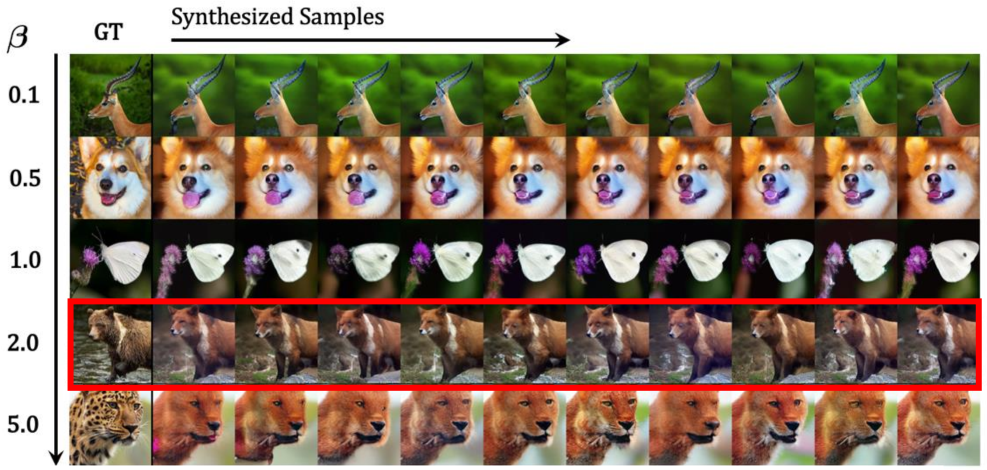
Linear Interpolation

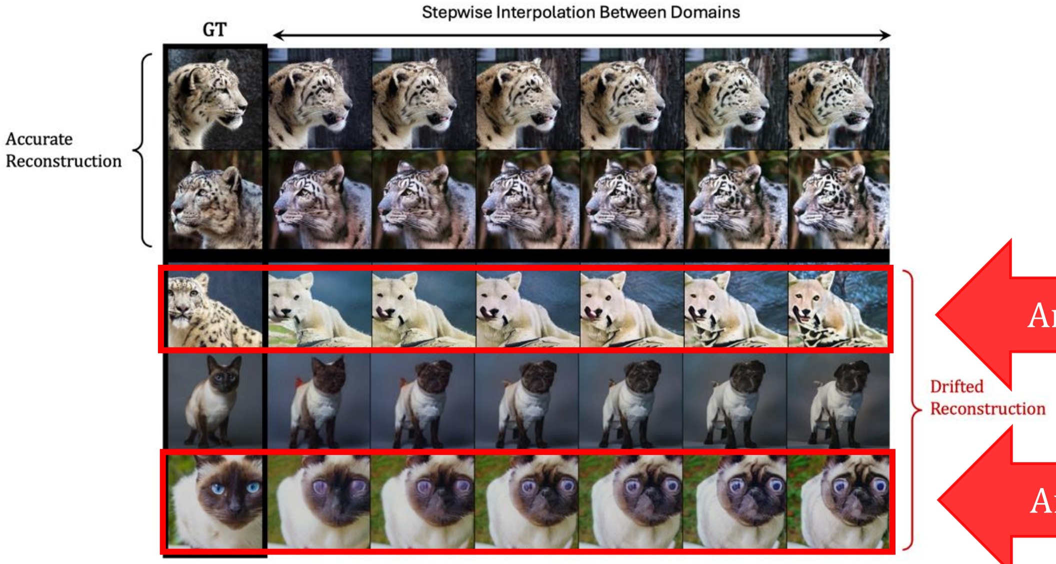
Linear Interpolation



Appendix: Limitations

Limitations





Self-Guidance Signal Ambiguity

- **Higher β-parameter** increases pressure while reducing capacity of the bottleneck layer, discarding both more **fine-grained details** and **class-specific information** – especially under high noise-levels

  - The high-level semantic information captured in the **self-guidance vector** may become **ambiguous** and cause **semantic mismatches** between ground-truth and the reconstructed high-frequency output image.
  - The **velocity decoder** may **drift** — producing incomplete outputs or samples from semantically related but incorrect classes *e.g., generating a fox or dog instead of a leopard, or a dog instead of a brown bear*.

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Appendix: Preliminary Analysis

Semantic Content Visualization

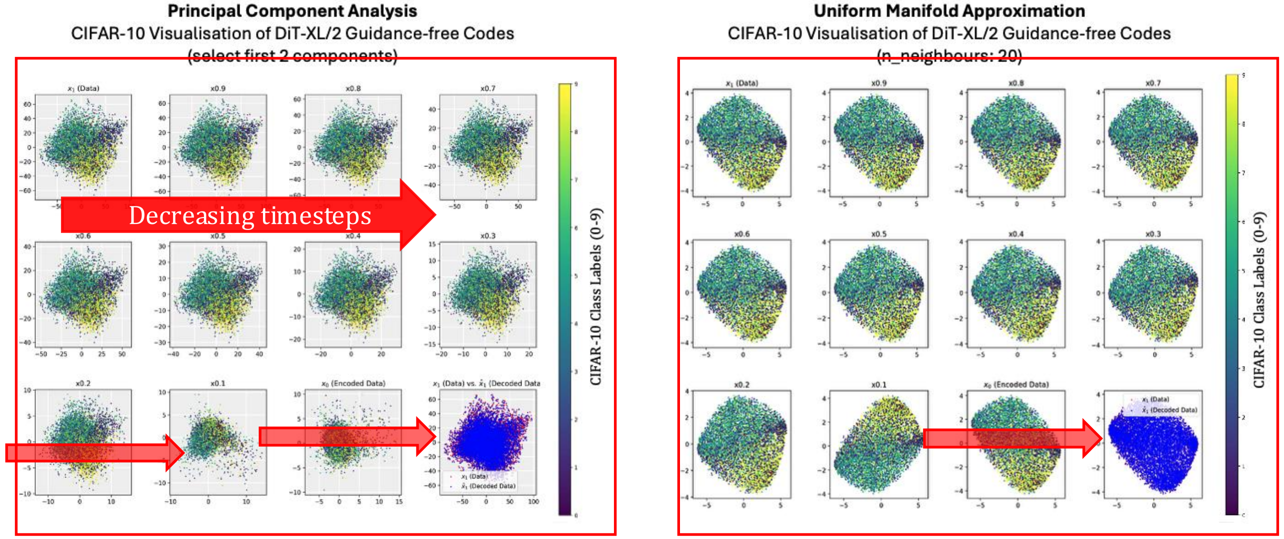

Figure: **Projection onto 2D space of learned latent noise samples.** Utilising dimensionality reduction techniques of PCA and UMAP for local and global structure analysis.

Dimensionality Reduction

- ODE sampling produces latent noises $\boldsymbol{x}_t$ which are **cycle consistency** (reconstruction in blue) [15].
- Latent variables $\boldsymbol{x}_t$ preserve **semantic meaning** along deterministic trajectories, even under high noise levels.
- Both non-linear UMAP and linear PCA reveal largely **stable, class-based clustering** across timesteps.
- UMAP clusters across timesteps show largely **homogeneous** distributions and clear class separation.
- Some linear structure and captured variance in PCA progressively fades out under high noise-levels masking meaningful image structure.

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Interpretability and Conditional Generation with Flow-Matching Models

- Presentation for the degree of M.Sc. in Computer Science
- Task assigned by: Prof. Dr. Björn Ommer
- Supervised by: Johannes Schusterbauer, M.Sc., PhD Student, Ming Gui, M.Sc., PhD Student
- Date of presentation: 09.09.2025